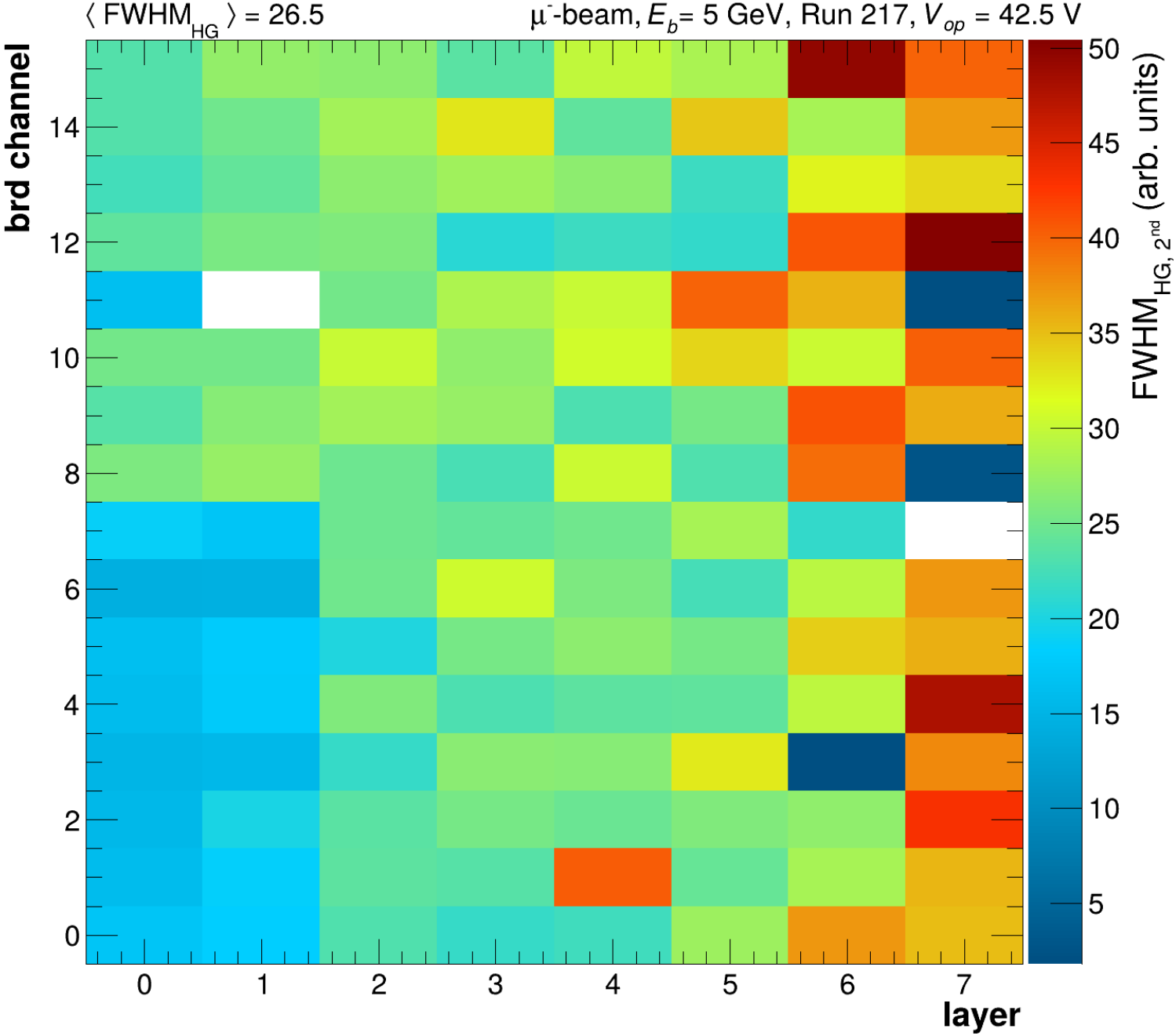
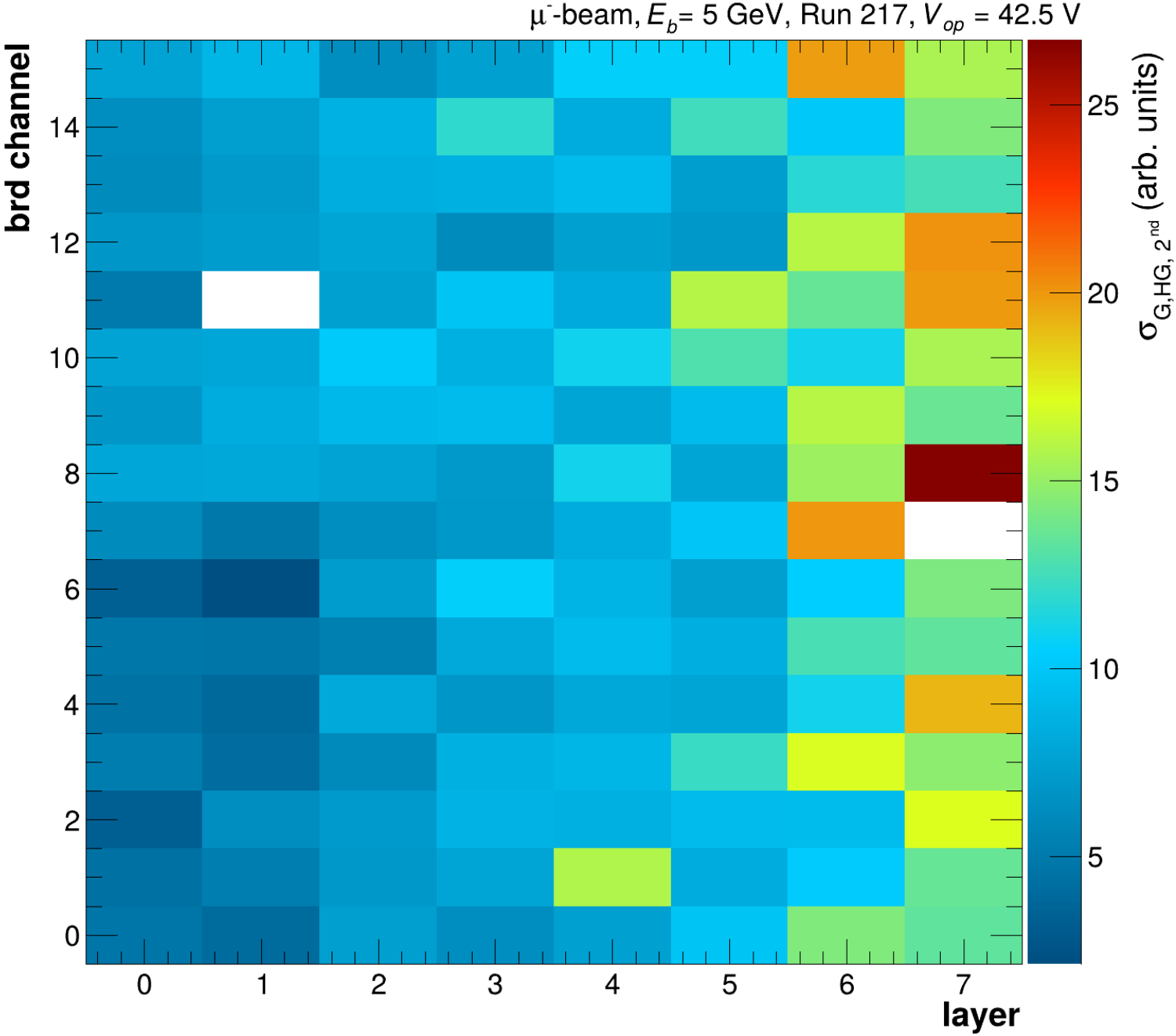
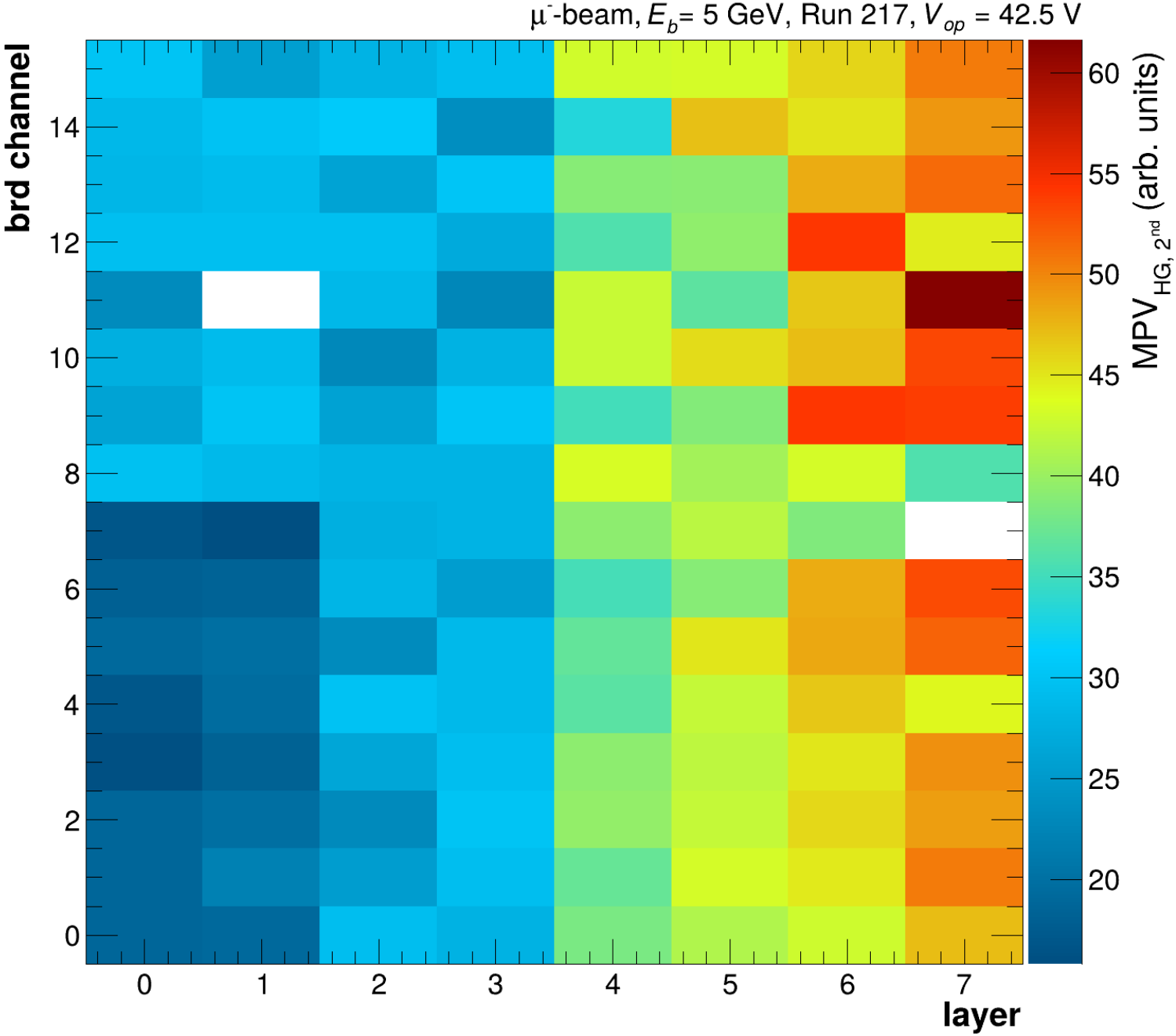
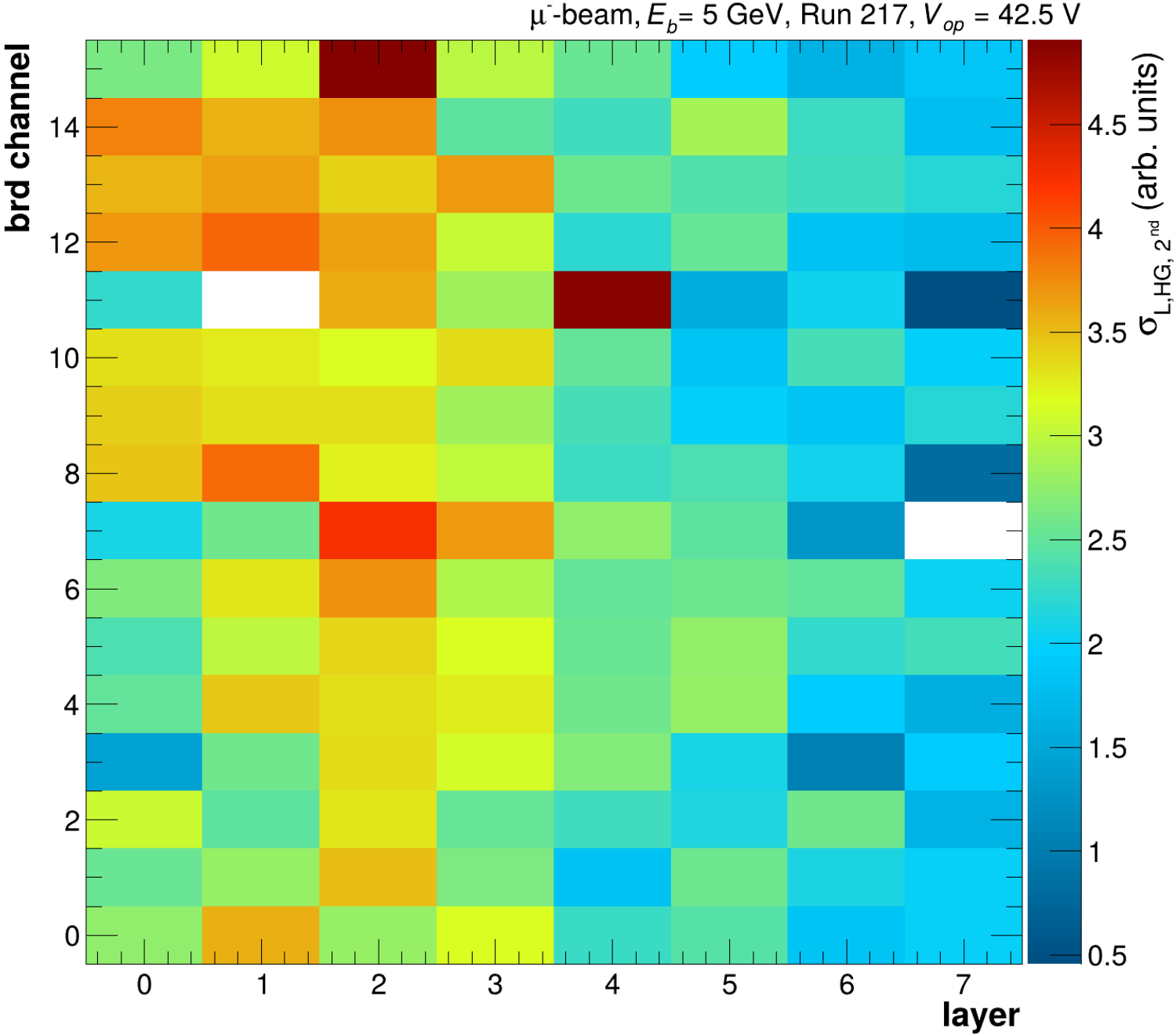
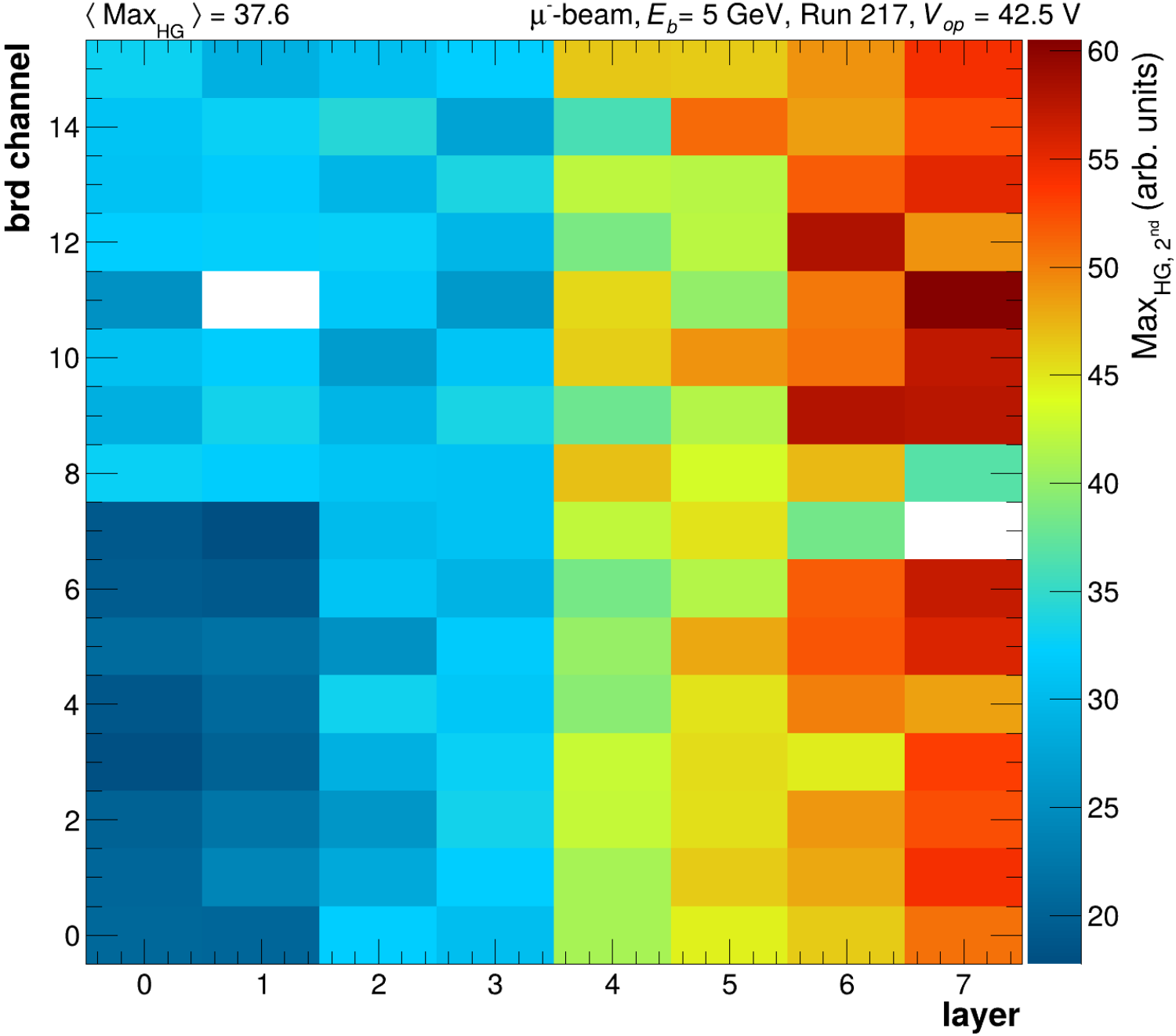
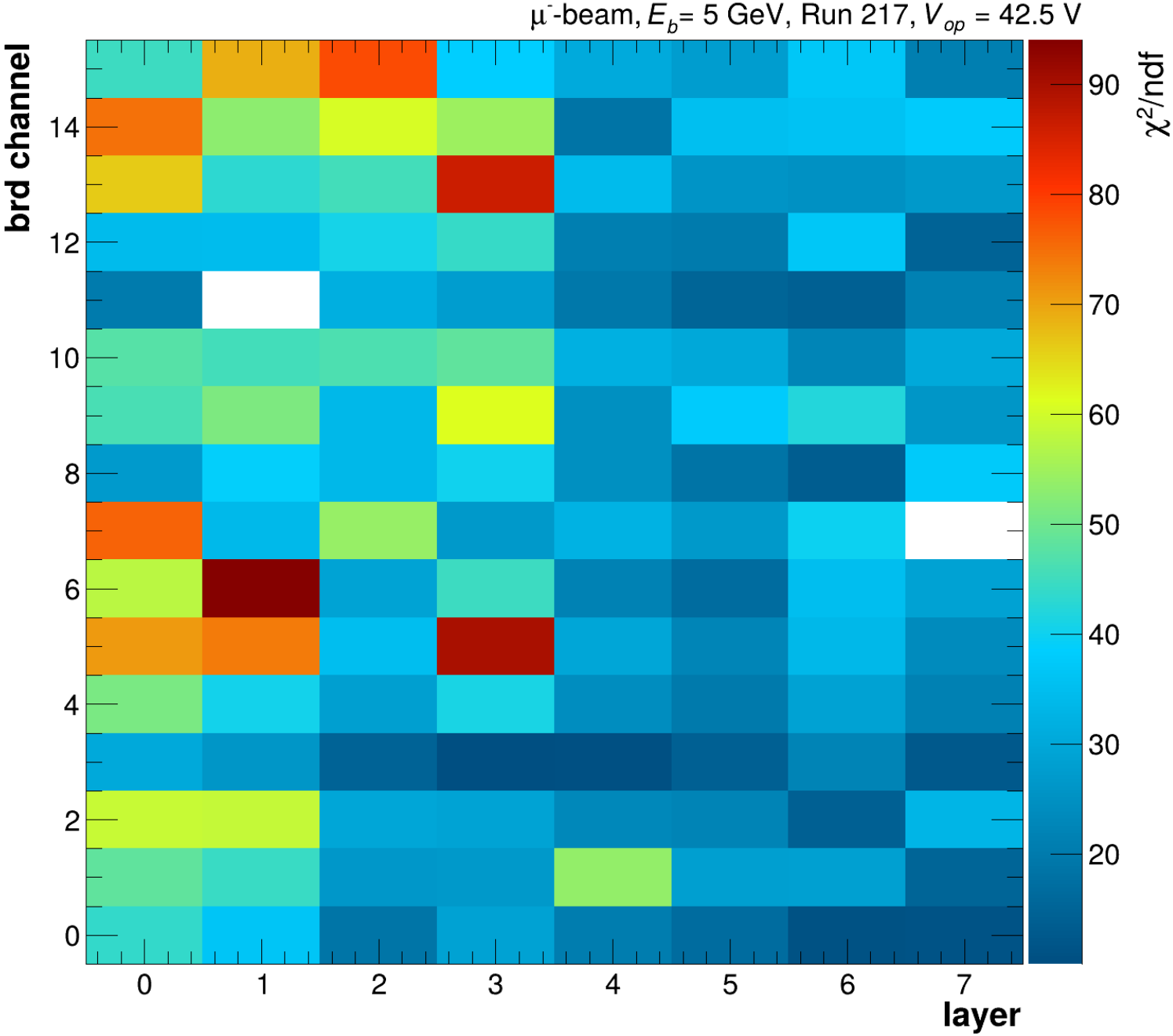

⟨ FWHM_HG ⟩ = 26.5
μ⁻-beam, E_b = 5 GeV, Run 217, V_op = 42.5 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35
40
45
50

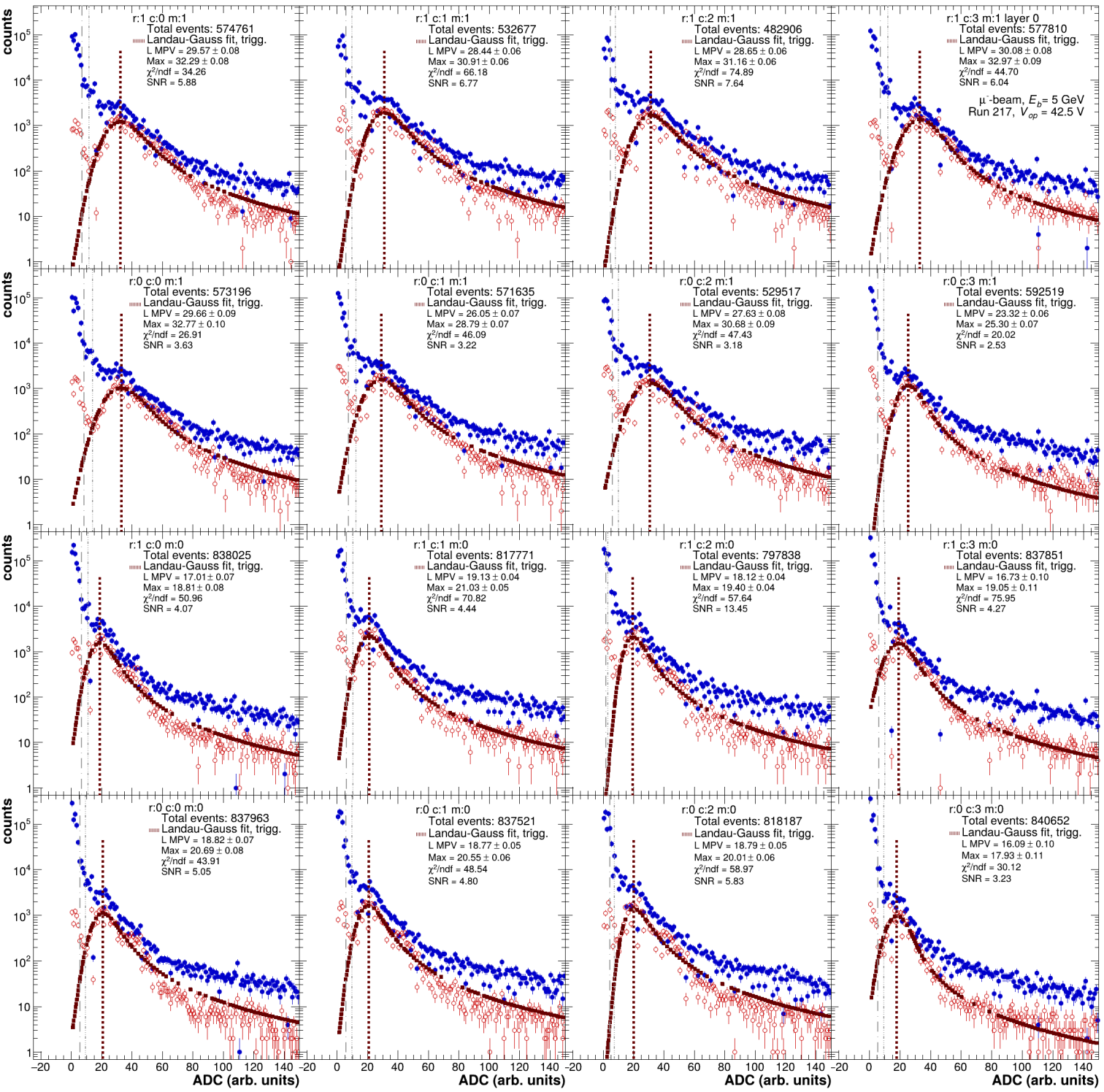

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25

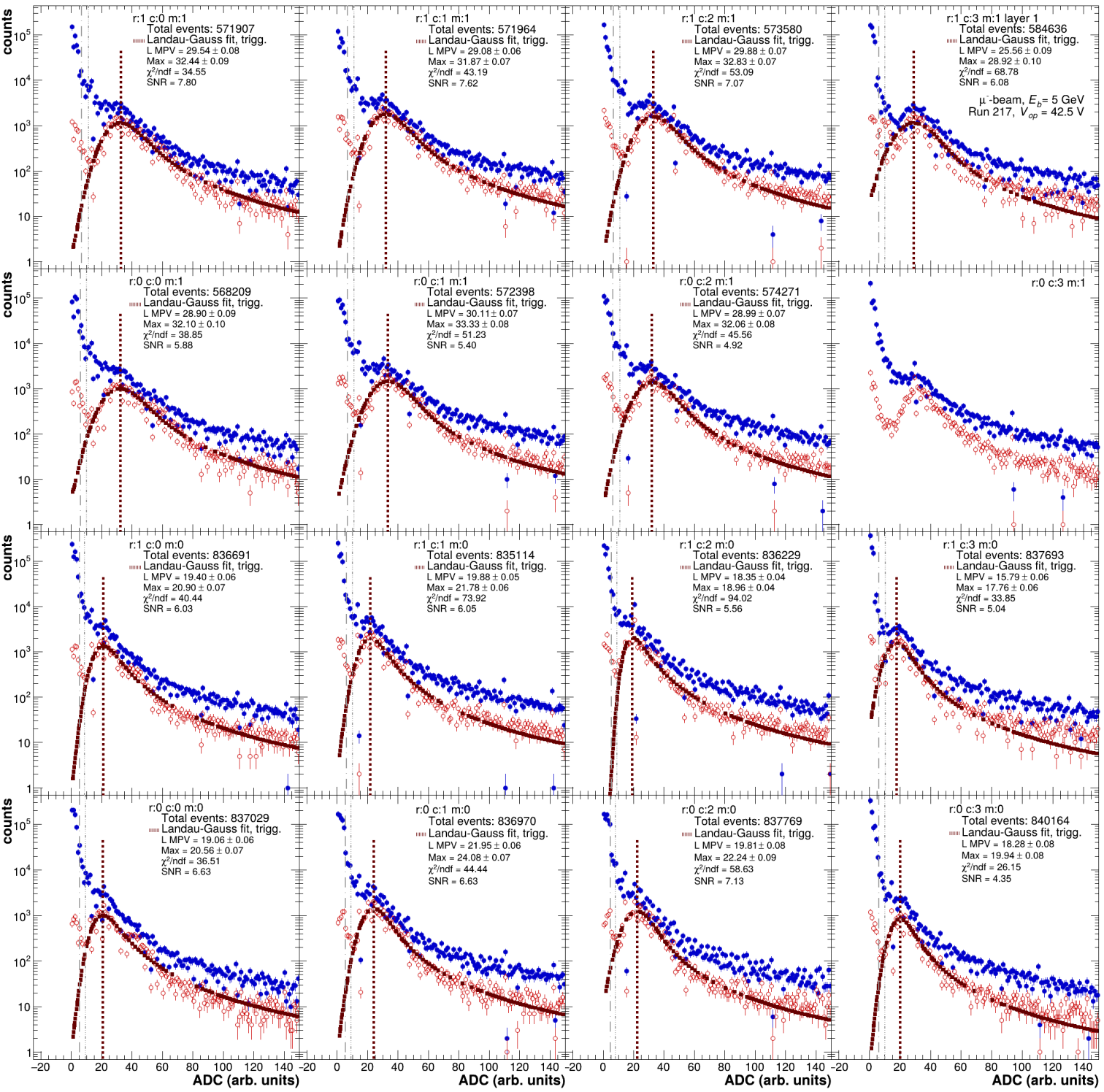

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
20
25
30
35
40
45
50
55
60
$MPV_{HG,\ 2^{nd}}$ (arb. units)

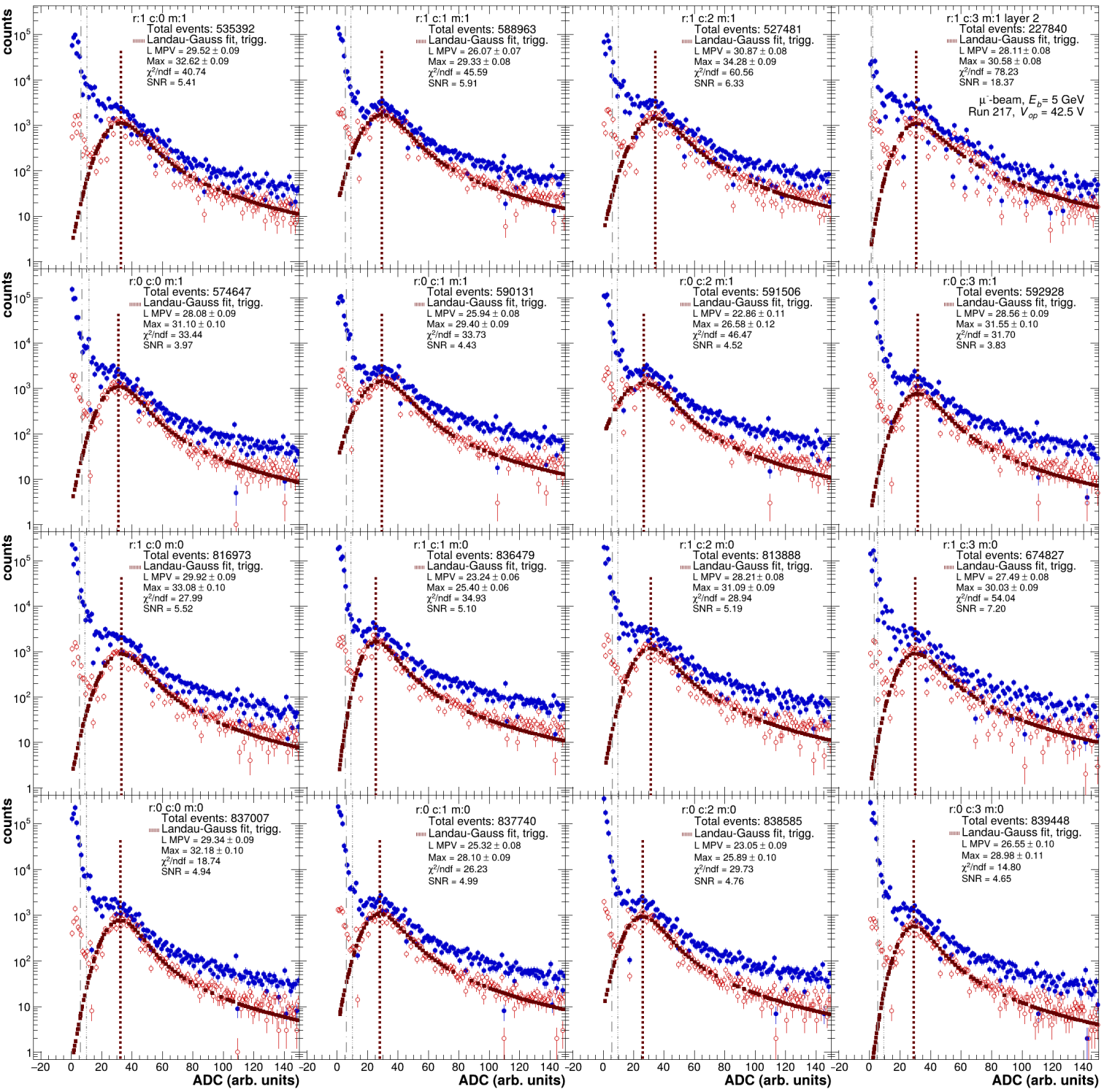

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0.5
1
1.5
2
2.5
3
3.5
4
4.5

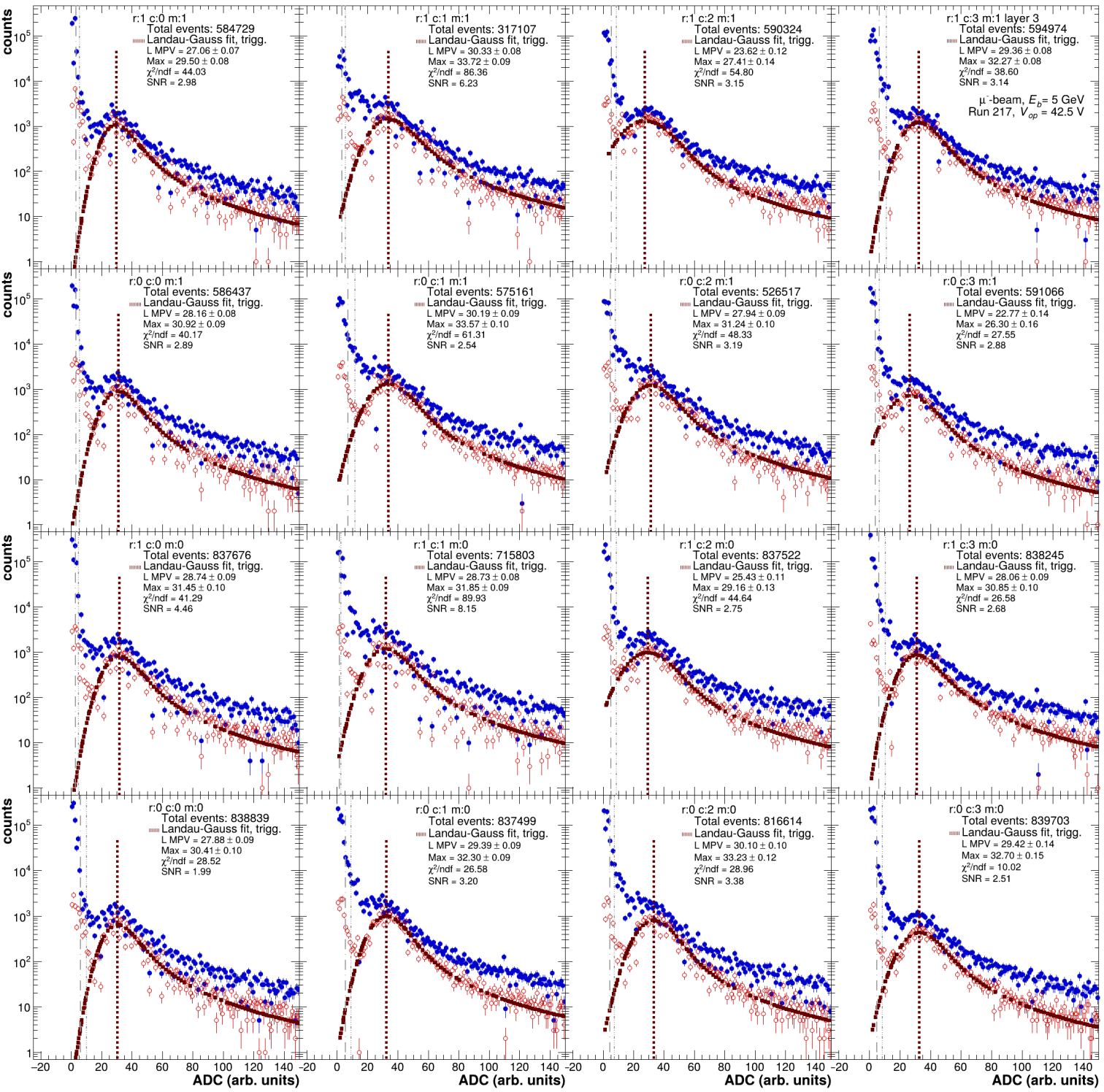

⟨ Max_HG ⟩ = 37.6
μ⁻-beam, E_b = 5 GeV, Run 217, V_op = 42.5 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
25
30
35
40
45
50
55
60

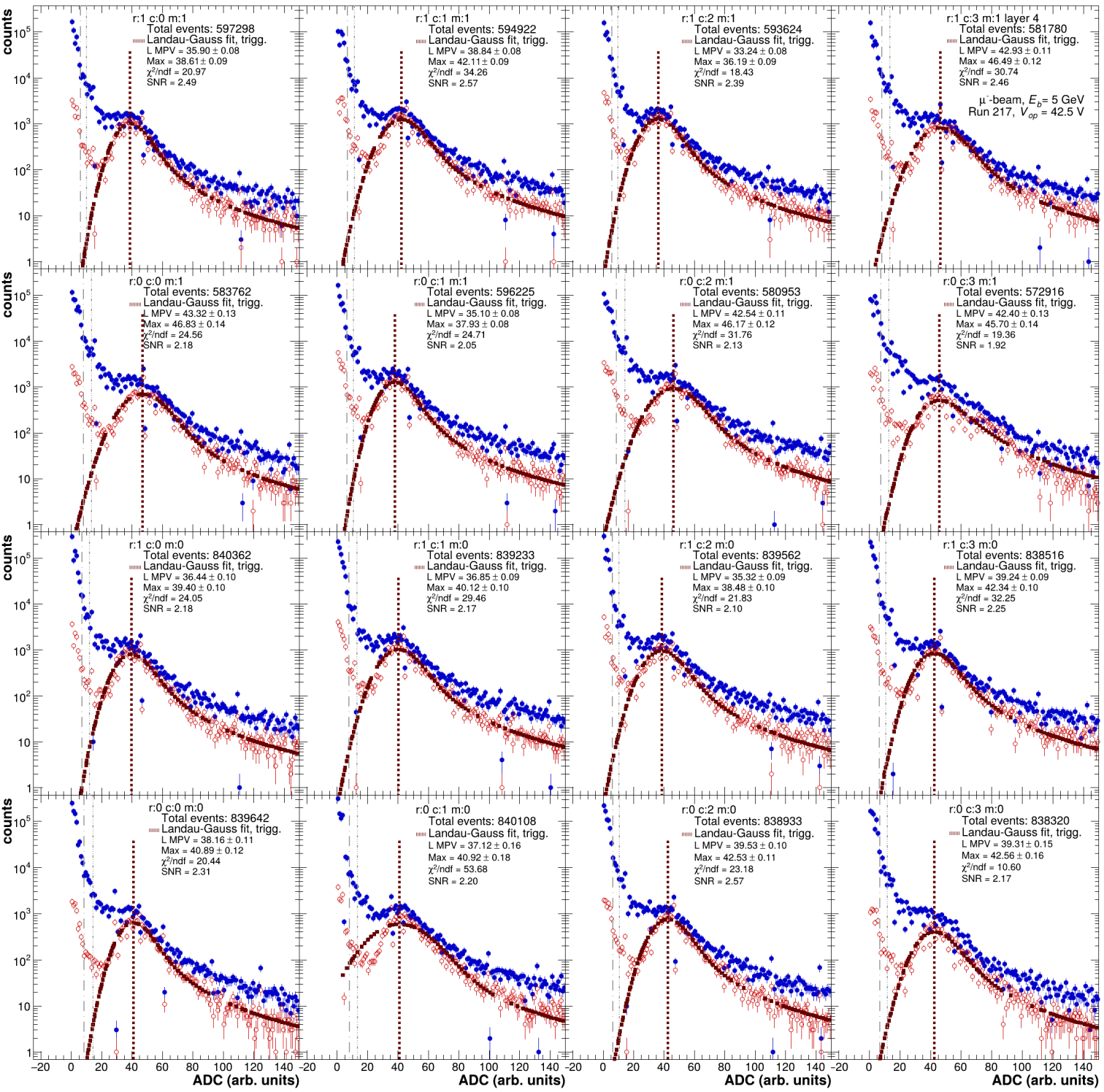

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
30
40
50
60
70
80
90

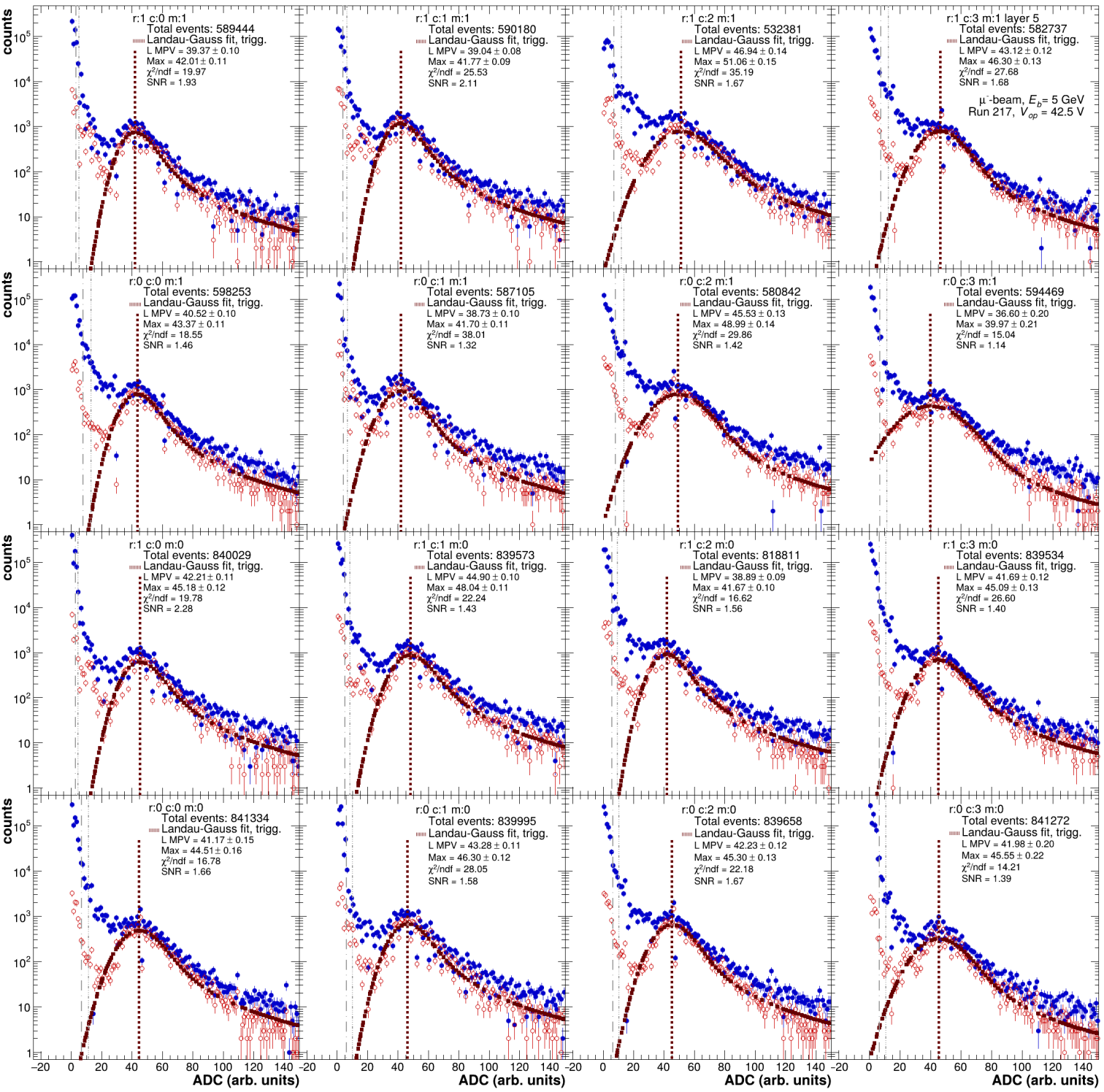

r:1 c:0 m:1
Total events: 574761
Landau-Gauss fit, trigg.
L MPV = 29.57 ± 0.08
Max = 32.29 ± 0.08
χ²/ndf = 34.26
SNR = 5.88
r:1 c:1 m:1
Total events: 532677
Landau-Gauss fit, trigg.
L MPV = 28.44 ± 0.06
Max = 30.91 ± 0.06
χ²/ndf = 66.18
SNR = 6.77
r:1 c:2 m:1
Total events: 482906
Landau-Gauss fit, trigg.
L MPV = 28.65 ± 0.06
Max = 31.16 ± 0.06
χ²/ndf = 74.89
SNR = 7.64
r:1 c:3 m:1 layer 0
Total events: 577810
Landau-Gauss fit, trigg.
L MPV = 30.08 ± 0.08
Max = 32.97 ± 0.09
χ²/ndf = 44.70
SNR = 6.04
μ⁻-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 573196
Landau-Gauss fit, trigg.
L MPV = 29.66 ± 0.09
Max = 32.77 ± 0.10
χ²/ndf = 26.91
SNR = 3.63
r:0 c:1 m:1
Total events: 571635
Landau-Gauss fit, trigg.
L MPV = 26.05 ± 0.07
Max = 28.79 ± 0.07
χ²/ndf = 46.09
SNR = 3.22
r:0 c:2 m:1
Total events: 529517
Landau-Gauss fit, trigg.
L MPV = 27.63 ± 0.08
Max = 30.68 ± 0.09
χ²/ndf = 47.43
SNR = 3.18
r:0 c:3 m:1
Total events: 592519
Landau-Gauss fit, trigg.
L MPV = 23.32 ± 0.06
Max = 25.30 ± 0.07
χ²/ndf = 20.02
SNR = 2.53
r:1 c:0 m:0
Total events: 838025
Landau-Gauss fit, trigg.
L MPV = 17.01 ± 0.07
Max = 18.81 ± 0.08
χ²/ndf = 50.96
SNR = 4.07
r:1 c:1 m:0
Total events: 817771
Landau-Gauss fit, trigg.
L MPV = 19.13 ± 0.04
Max = 21.03 ± 0.05
χ²/ndf = 70.82
SNR = 4.44
r:1 c:2 m:0
Total events: 797838
Landau-Gauss fit, trigg.
L MPV = 18.12 ± 0.04
Max = 19.40 ± 0.04
χ²/ndf = 57.64
SNR = 13.45
r:1 c:3 m:0
Total events: 837851
Landau-Gauss fit, trigg.
L MPV = 16.73 ± 0.10
Max = 19.05 ± 0.11
χ²/ndf = 75.95
SNR = 4.27
r:0 c:0 m:0
Total events: 837963
Landau-Gauss fit, trigg.
L MPV = 18.82 ± 0.07
Max = 20.69 ± 0.08
χ²/ndf = 43.91
SNR = 5.05
r:0 c:1 m:0
Total events: 837521
Landau-Gauss fit, trigg.
L MPV = 18.77 ± 0.05
Max = 20.55 ± 0.06
χ²/ndf = 48.54
SNR = 4.80
r:0 c:2 m:0
Total events: 818187
Landau-Gauss fit, trigg.
L MPV = 18.79 ± 0.05
Max = 20.01 ± 0.06
χ²/ndf = 58.97
SNR = 5.83
r:0 c:3 m:0
Total events: 840652
Landau-Gauss fit, trigg.
L MPV = 16.09 ± 0.10
Max = 17.93 ± 0.11
χ²/ndf = 30.12
SNR = 3.23
counts
ADC (arb. units)

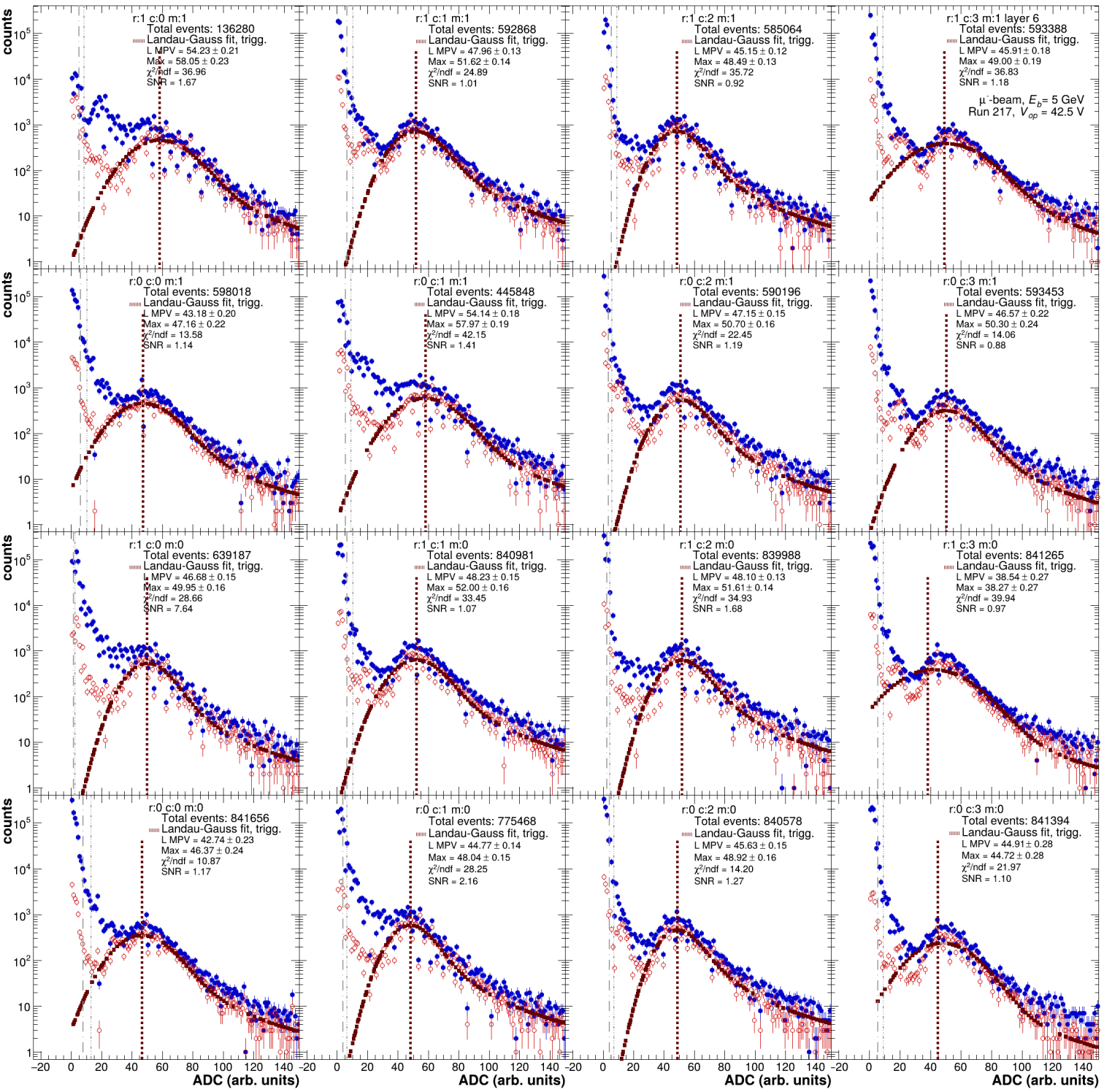

r:1 c:0 m:1
Total events: 571907
Landau-Gauss fit, trigg.
L MPV = 29.54 ± 0.08
Max = 32.44 ± 0.09
χ²/ndf = 34.55
SNR = 7.80
r:1 c:1 m:1
Total events: 571964
Landau-Gauss fit, trigg.
L MPV = 29.08 ± 0.06
Max = 31.87 ± 0.07
χ²/ndf = 43.19
SNR = 7.62
r:1 c:2 m:1
Total events: 573580
Landau-Gauss fit, trigg.
L MPV = 29.88 ± 0.07
Max = 32.83 ± 0.07
χ²/ndf = 53.09
SNR = 7.07
r:1 c:3 m:1 layer 1
Total events: 584636
Landau-Gauss fit, trigg.
L MPV = 25.56 ± 0.09
Max = 28.92 ± 0.10
χ²/ndf = 68.78
SNR = 6.08
μ⁻-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 568209
Landau-Gauss fit, trigg.
L MPV = 28.90 ± 0.09
Max = 32.10 ± 0.10
χ²/ndf = 38.85
SNR = 5.88
r:0 c:1 m:1
Total events: 572398
Landau-Gauss fit, trigg.
L MPV = 30.11 ± 0.07
Max = 33.33 ± 0.08
χ²/ndf = 51.23
SNR = 5.40
r:0 c:2 m:1
Total events: 574271
Landau-Gauss fit, trigg.
L MPV = 28.99 ± 0.07
Max = 32.06 ± 0.08
χ²/ndf = 45.56
SNR = 4.92
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 836691
Landau-Gauss fit, trigg.
L MPV = 19.40 ± 0.06
Max = 20.90 ± 0.07
χ²/ndf = 40.44
SNR = 6.03
r:1 c:1 m:0
Total events: 835114
Landau-Gauss fit, trigg.
L MPV = 19.88 ± 0.05
Max = 21.78 ± 0.06
χ²/ndf = 73.92
SNR = 6.05
r:1 c:2 m:0
Total events: 836229
Landau-Gauss fit, trigg.
L MPV = 18.35 ± 0.04
Max = 18.96 ± 0.04
χ²/ndf = 94.02
SNR = 5.56
r:1 c:3 m:0
Total events: 837693
Landau-Gauss fit, trigg.
L MPV = 15.79 ± 0.06
Max = 17.76 ± 0.06
χ²/ndf = 33.85
SNR = 5.04
r:0 c:0 m:0
Total events: 837029
Landau-Gauss fit, trigg.
L MPV = 19.06 ± 0.06
Max = 20.56 ± 0.07
χ²/ndf = 36.51
SNR = 6.63
r:0 c:1 m:0
Total events: 836970
Landau-Gauss fit, trigg.
L MPV = 21.95 ± 0.06
Max = 24.08 ± 0.07
χ²/ndf = 44.44
SNR = 6.63
r:0 c:2 m:0
Total events: 837769
Landau-Gauss fit, trigg.
L MPV = 19.81 ± 0.08
Max = 22.24 ± 0.09
χ²/ndf = 58.63
SNR = 7.13
r:0 c:3 m:0
Total events: 840164
Landau-Gauss fit, trigg.
L MPV = 18.28 ± 0.08
Max = 19.94 ± 0.08
χ²/ndf = 26.15
SNR = 4.35
counts
ADC (arb. units)

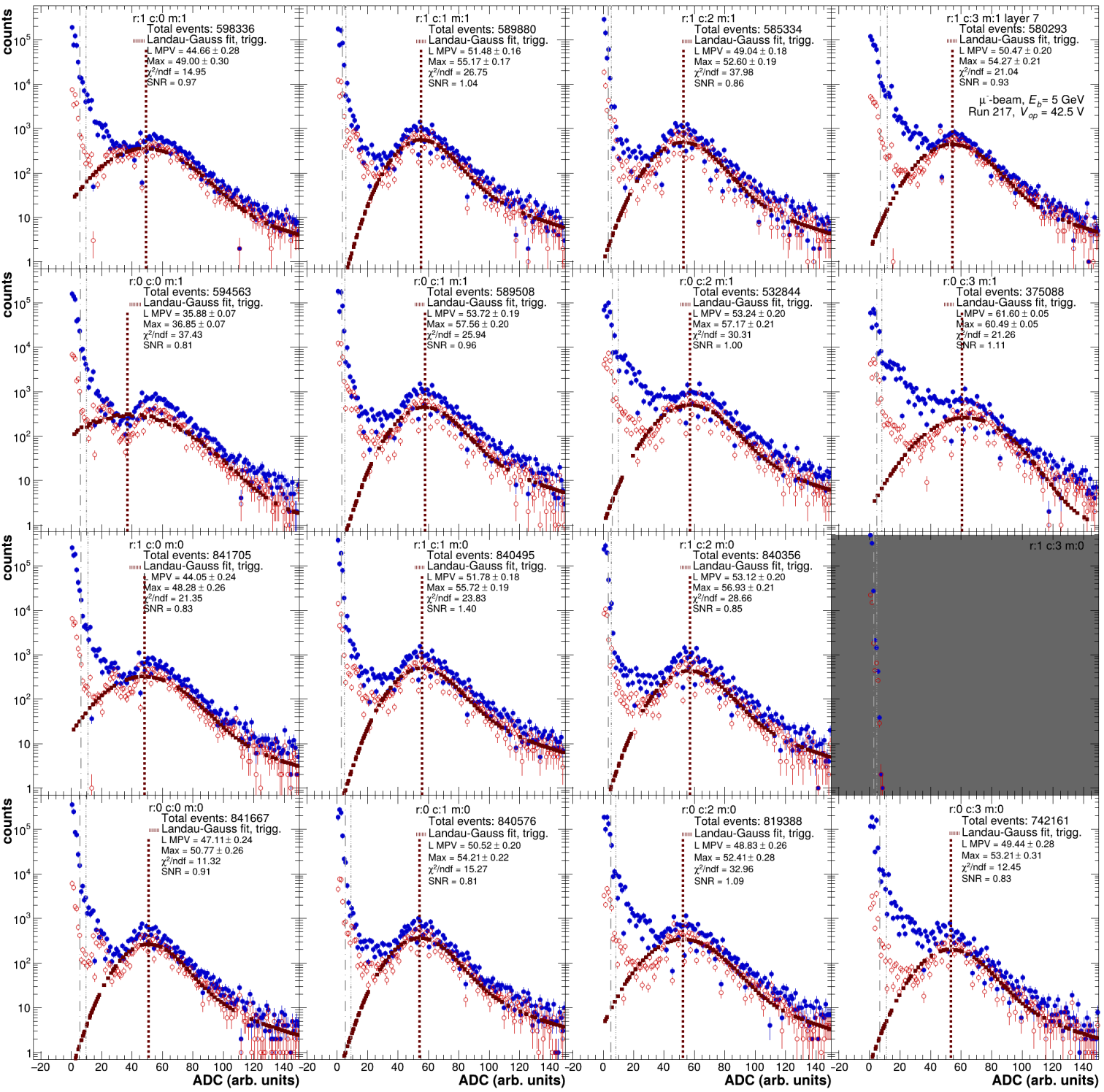

r:1 c:0 m:1
Total events: 535392
Landau-Gauss fit, trigg.
L MPV = 29.52 ± 0.09
Max = 32.62 ± 0.09
χ²/ndf = 40.74
SNR = 5.41
r:1 c:1 m:1
Total events: 588963
Landau-Gauss fit, trigg.
L MPV = 26.07 ± 0.07
Max = 29.33 ± 0.08
χ²/ndf = 45.59
SNR = 5.91
r:1 c:2 m:1
Total events: 527481
Landau-Gauss fit, trigg.
L MPV = 30.87 ± 0.08
Max = 34.28 ± 0.09
χ²/ndf = 60.56
SNR = 6.33
r:1 c:3 m:1 layer 2
Total events: 227840
Landau-Gauss fit, trigg.
L MPV = 28.11 ± 0.08
Max = 30.58 ± 0.08
χ²/ndf = 78.23
SNR = 18.37
μ⁻-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 574647
Landau-Gauss fit, trigg.
L MPV = 28.08 ± 0.09
Max = 31.10 ± 0.10
χ²/ndf = 33.44
SNR = 3.97
r:0 c:1 m:1
Total events: 590131
Landau-Gauss fit, trigg.
L MPV = 25.94 ± 0.08
Max = 29.40 ± 0.09
χ²/ndf = 33.73
SNR = 4.43
r:0 c:2 m:1
Total events: 591506
Landau-Gauss fit, trigg.
L MPV = 22.86 ± 0.11
Max = 26.58 ± 0.12
χ²/ndf = 46.47
SNR = 4.52
r:0 c:3 m:1
Total events: 592928
Landau-Gauss fit, trigg.
L MPV = 28.56 ± 0.09
Max = 31.55 ± 0.10
χ²/ndf = 31.70
SNR = 3.83
r:1 c:0 m:0
Total events: 816973
Landau-Gauss fit, trigg.
L MPV = 29.92 ± 0.09
Max = 33.08 ± 0.10
χ²/ndf = 27.99
SNR = 5.52
r:1 c:1 m:0
Total events: 836479
Landau-Gauss fit, trigg.
L MPV = 23.24 ± 0.06
Max = 25.40 ± 0.06
χ²/ndf = 34.93
SNR = 5.10
r:1 c:2 m:0
Total events: 813888
Landau-Gauss fit, trigg.
L MPV = 28.21 ± 0.08
Max = 31.09 ± 0.09
χ²/ndf = 28.94
SNR = 5.19
r:1 c:3 m:0
Total events: 674827
Landau-Gauss fit, trigg.
L MPV = 27.49 ± 0.08
Max = 30.03 ± 0.09
χ²/ndf = 54.04
SNR = 7.20
r:0 c:0 m:0
Total events: 837007
Landau-Gauss fit, trigg.
L MPV = 29.34 ± 0.09
Max = 32.18 ± 0.10
χ²/ndf = 18.74
SNR = 4.94
r:0 c:1 m:0
Total events: 837740
Landau-Gauss fit, trigg.
L MPV = 25.32 ± 0.08
Max = 28.10 ± 0.09
χ²/ndf = 26.23
SNR = 4.99
r:0 c:2 m:0
Total events: 838585
Landau-Gauss fit, trigg.
L MPV = 23.05 ± 0.09
Max = 25.89 ± 0.10
χ²/ndf = 29.73
SNR = 4.76
r:0 c:3 m:0
Total events: 839448
Landau-Gauss fit, trigg.
L MPV = 26.55 ± 0.10
Max = 28.98 ± 0.11
χ²/ndf = 14.80
SNR = 4.65
counts
ADC (arb. units)

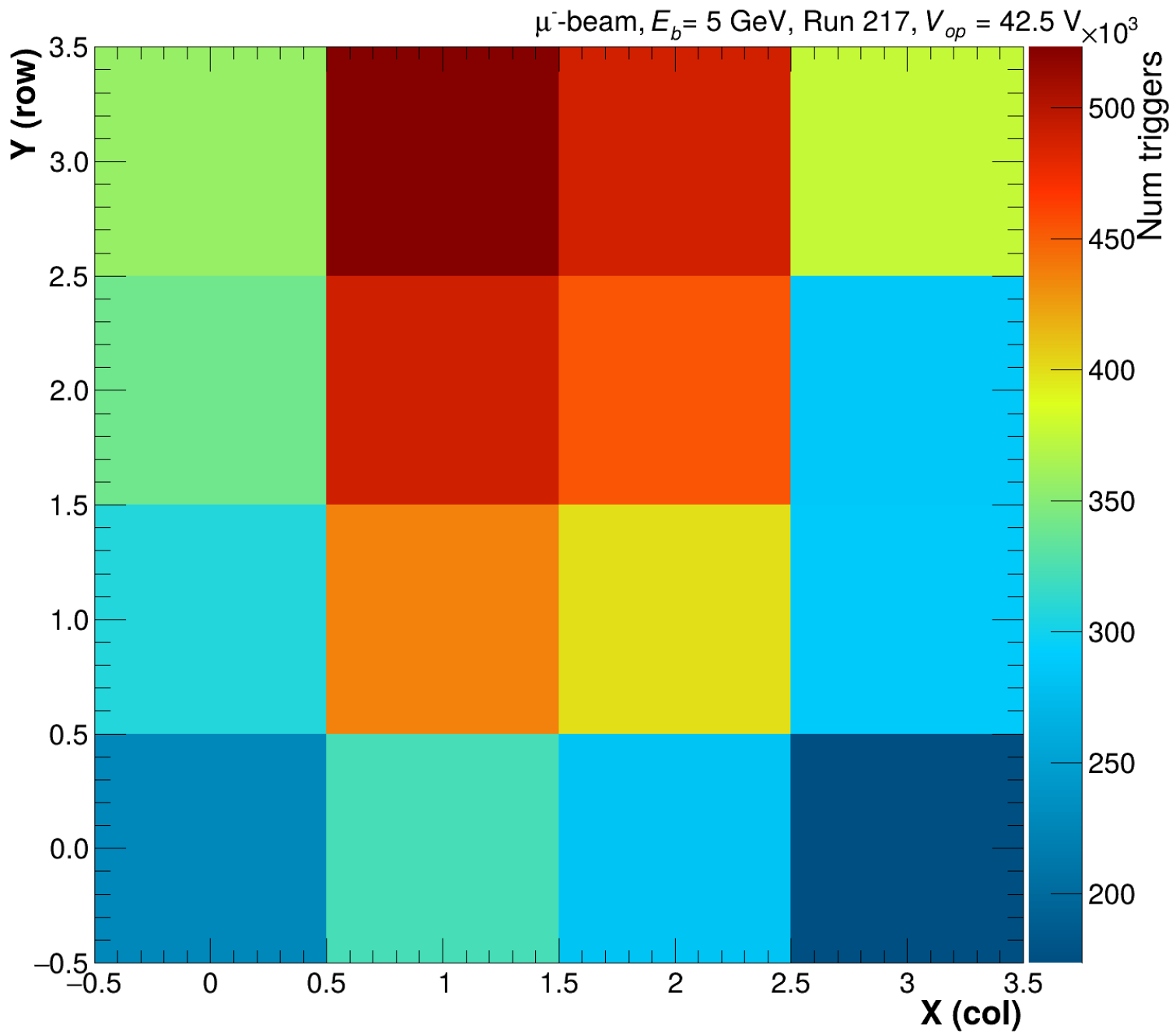

r:1 c:0 m:1
Total events: 584729
Landau-Gauss fit, trigg.
L MPV = 27.06 ± 0.07
Max = 29.50 ± 0.08
χ²/ndf = 44.03
SNR = 2.98
r:1 c:1 m:1
Total events: 317107
Landau-Gauss fit, trigg.
L MPV = 30.33 ± 0.08
Max = 33.72 ± 0.09
χ²/ndf = 86.36
SNR = 6.23
r:1 c:2 m:1
Total events: 590324
Landau-Gauss fit, trigg.
L MPV = 23.62 ± 0.12
Max = 27.41 ± 0.14
χ²/ndf = 54.80
SNR = 3.15
r:1 c:3 m:1 layer 3
Total events: 594974
Landau-Gauss fit, trigg.
L MPV = 29.36 ± 0.08
Max = 32.27 ± 0.08
χ²/ndf = 38.60
SNR = 3.14
μ-beam, E_b = 5 GeV
Run 217, V_{op} = 42.5 V
r:0 c:0 m:1
Total events: 586437
Landau-Gauss fit, trigg.
L MPV = 28.16 ± 0.08
Max = 30.92 ± 0.09
χ²/ndf = 40.17
SNR = 2.89
r:0 c:1 m:1
Total events: 575161
Landau-Gauss fit, trigg.
L MPV = 30.19 ± 0.09
Max = 33.57 ± 0.10
χ²/ndf = 61.31
SNR = 2.54
r:0 c:2 m:1
Total events: 526517
Landau-Gauss fit, trigg.
L MPV = 27.94 ± 0.09
Max = 31.24 ± 0.10
χ²/ndf = 48.33
SNR = 3.19
r:0 c:3 m:1
Total events: 591066
Landau-Gauss fit, trigg.
L MPV = 22.77 ± 0.14
Max = 26.30 ± 0.16
χ²/ndf = 27.55
SNR = 2.88
r:1 c:0 m:0
Total events: 837676
Landau-Gauss fit, trigg.
L MPV = 28.74 ± 0.09
Max = 31.45 ± 0.10
χ²/ndf = 41.29
SNR = 4.46
r:1 c:1 m:0
Total events: 715803
Landau-Gauss fit, trigg.
L MPV = 28.73 ± 0.08
Max = 31.85 ± 0.09
χ²/ndf = 89.93
SNR = 8.15
r:1 c:2 m:0
Total events: 837522
Landau-Gauss fit, trigg.
L MPV = 25.43 ± 0.11
Max = 29.16 ± 0.13
χ²/ndf = 44.64
SNR = 2.75
r:1 c:3 m:0
Total events: 838245
Landau-Gauss fit, trigg.
L MPV = 28.06 ± 0.09
Max = 30.85 ± 0.10
χ²/ndf = 26.58
SNR = 2.68
r:0 c:0 m:0
Total events: 838839
Landau-Gauss fit, trigg.
L MPV = 27.88 ± 0.09
Max = 30.41 ± 0.10
χ²/ndf = 28.52
SNR = 1.99
r:0 c:1 m:0
Total events: 837499
Landau-Gauss fit, trigg.
L MPV = 29.39 ± 0.09
Max = 32.30 ± 0.09
χ²/ndf = 26.58
SNR = 3.20
r:0 c:2 m:0
Total events: 816614
Landau-Gauss fit, trigg.
L MPV = 30.10 ± 0.10
Max = 33.23 ± 0.12
χ²/ndf = 28.96
SNR = 3.38
r:0 c:3 m:0
Total events: 839703
Landau-Gauss fit, trigg.
L MPV = 29.42 ± 0.14
Max = 32.70 ± 0.15
χ²/ndf = 10.02
SNR = 2.51
counts
ADC (arb. units)

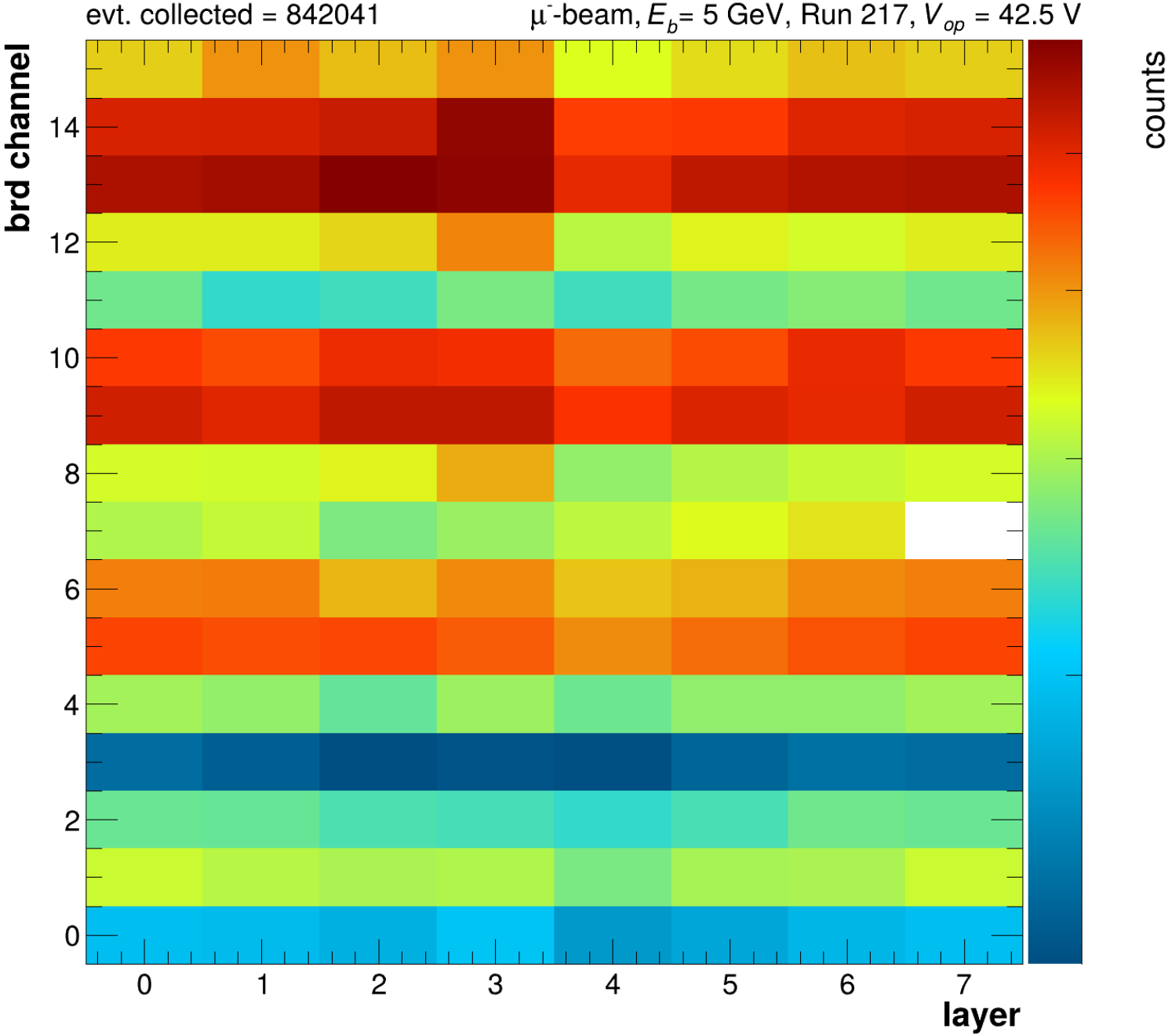

r:1 c:0 m:1
Total events: 597298
Landau-Gauss fit, trigg.
L MPV = 35.90 ± 0.08
Max = 38.61 ± 0.09
χ²/ndf = 20.97
SNR = 2.49
r:1 c:1 m:1
Total events: 594922
Landau-Gauss fit, trigg.
L MPV = 38.84 ± 0.08
Max = 42.11 ± 0.09
χ²/ndf = 34.26
SNR = 2.57
r:1 c:2 m:1
Total events: 593624
Landau-Gauss fit, trigg.
L MPV = 33.24 ± 0.08
Max = 36.19 ± 0.09
χ²/ndf = 18.43
SNR = 2.39
r:1 c:3 m:1 layer 4
Total events: 581780
Landau-Gauss fit, trigg.
L MPV = 42.93 ± 0.11
Max = 46.49 ± 0.12
χ²/ndf = 30.74
SNR = 2.46
μ-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 583762
Landau-Gauss fit, trigg.
L MPV = 43.32 ± 0.13
Max = 46.83 ± 0.14
χ²/ndf = 24.56
SNR = 2.18
r:0 c:1 m:1
Total events: 596225
Landau-Gauss fit, trigg.
L MPV = 35.10 ± 0.08
Max = 37.93 ± 0.08
χ²/ndf = 24.71
SNR = 2.05
r:0 c:2 m:1
Total events: 580953
Landau-Gauss fit, trigg.
L MPV = 42.54 ± 0.11
Max = 46.17 ± 0.12
χ²/ndf = 31.76
SNR = 2.13
r:0 c:3 m:1
Total events: 572916
Landau-Gauss fit, trigg.
L MPV = 42.40 ± 0.13
Max = 45.70 ± 0.14
χ²/ndf = 19.36
SNR = 1.92
r:1 c:0 m:0
Total events: 840362
Landau-Gauss fit, trigg.
L MPV = 36.44 ± 0.10
Max = 39.40 ± 0.10
χ²/ndf = 24.05
SNR = 2.18
r:1 c:1 m:0
Total events: 839233
Landau-Gauss fit, trigg.
L MPV = 36.85 ± 0.09
Max = 40.12 ± 0.10
χ²/ndf = 29.46
SNR = 2.17
r:1 c:2 m:0
Total events: 839562
Landau-Gauss fit, trigg.
L MPV = 35.32 ± 0.09
Max = 38.48 ± 0.10
χ²/ndf = 21.83
SNR = 2.10
r:1 c:3 m:0
Total events: 838516
Landau-Gauss fit, trigg.
L MPV = 39.24 ± 0.09
Max = 42.34 ± 0.10
χ²/ndf = 32.25
SNR = 2.25
r:0 c:0 m:0
Total events: 839642
Landau-Gauss fit, trigg.
L MPV = 38.16 ± 0.11
Max = 40.89 ± 0.12
χ²/ndf = 20.44
SNR = 2.31
r:0 c:1 m:0
Total events: 840108
Landau-Gauss fit, trigg.
L MPV = 37.12 ± 0.16
Max = 40.92 ± 0.18
χ²/ndf = 53.68
SNR = 2.20
r:0 c:2 m:0
Total events: 838933
Landau-Gauss fit, trigg.
L MPV = 39.53 ± 0.10
Max = 42.53 ± 0.11
χ²/ndf = 23.18
SNR = 2.57
r:0 c:3 m:0
Total events: 838320
Landau-Gauss fit, trigg.
L MPV = 39.31 ± 0.15
Max = 42.56 ± 0.16
χ²/ndf = 10.60
SNR = 2.17
counts
ADC (arb. units)

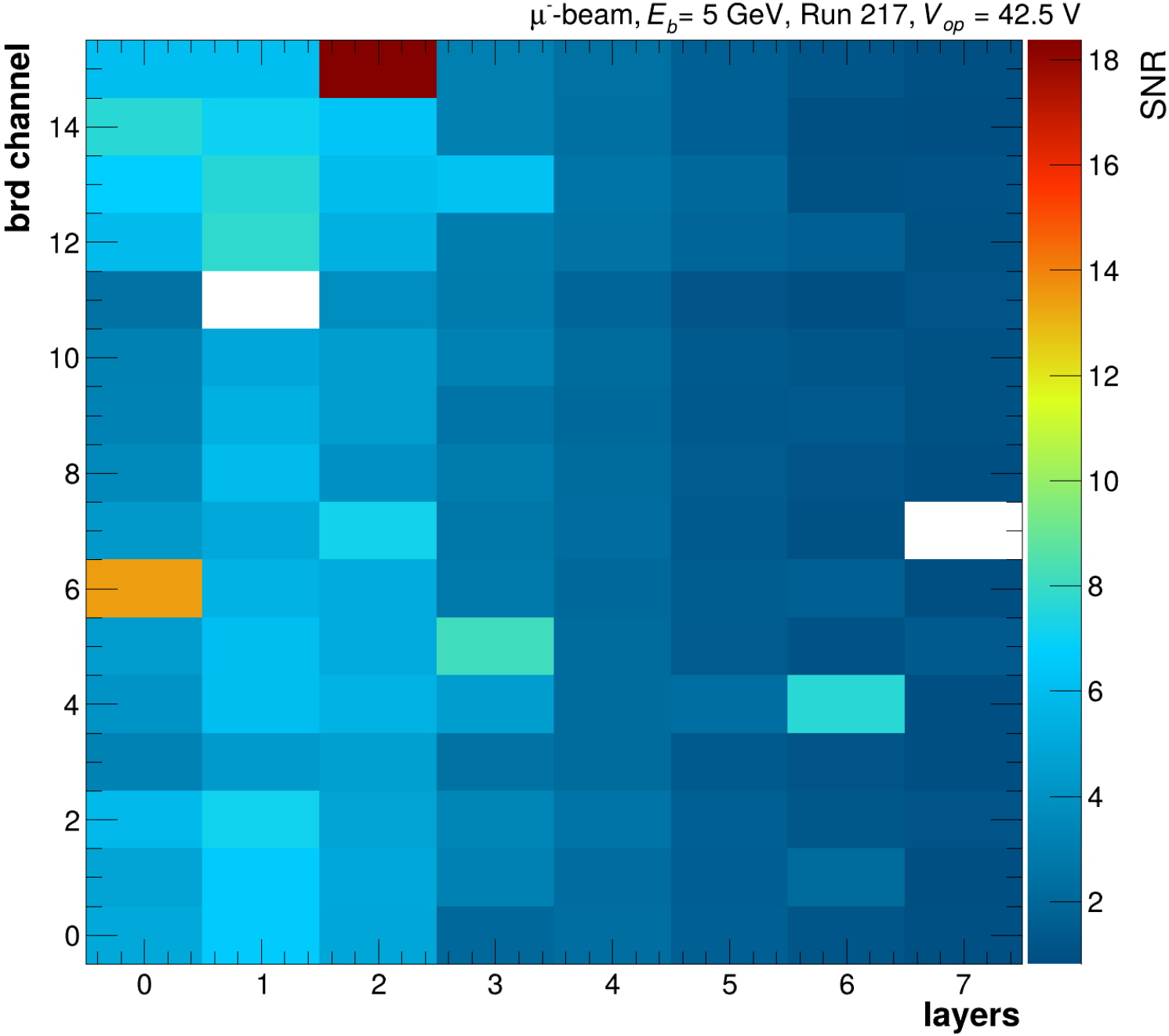

r:1 c:0 m:1
Total events: 589444
Landau-Gauss fit, trigg.
L MPV = 39.37 ± 0.10
Max = 42.01 ± 0.11
χ²/ndf = 19.97
SNR = 1.93
r:1 c:1 m:1
Total events: 590180
Landau-Gauss fit, trigg.
L MPV = 39.04 ± 0.08
Max = 41.77 ± 0.09
χ²/ndf = 25.53
SNR = 2.11
r:1 c:2 m:1
Total events: 532381
Landau-Gauss fit, trigg.
L MPV = 46.94 ± 0.14
Max = 51.06 ± 0.15
χ²/ndf = 35.19
SNR = 1.67
r:1 c:3 m:1 layer 5
Total events: 582737
Landau-Gauss fit, trigg.
L MPV = 43.12 ± 0.12
Max = 46.30 ± 0.13
χ²/ndf = 27.68
SNR = 1.68
μ⁻-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 598253
Landau-Gauss fit, trigg.
L MPV = 40.52 ± 0.10
Max = 43.37 ± 0.11
χ²/ndf = 18.55
SNR = 1.46
r:0 c:1 m:1
Total events: 587105
Landau-Gauss fit, trigg.
L MPV = 38.73 ± 0.10
Max = 41.70 ± 0.11
χ²/ndf = 38.01
SNR = 1.32
r:0 c:2 m:1
Total events: 580842
Landau-Gauss fit, trigg.
L MPV = 45.53 ± 0.13
Max = 48.99 ± 0.14
χ²/ndf = 29.86
SNR = 1.42
r:0 c:3 m:1
Total events: 594469
Landau-Gauss fit, trigg.
L MPV = 36.60 ± 0.20
Max = 39.97 ± 0.21
χ²/ndf = 15.04
SNR = 1.14
r:1 c:0 m:0
Total events: 840029
Landau-Gauss fit, trigg.
L MPV = 42.21 ± 0.11
Max = 45.18 ± 0.12
χ²/ndf = 19.78
SNR = 2.28
r:1 c:1 m:0
Total events: 839573
Landau-Gauss fit, trigg.
L MPV = 44.90 ± 0.10
Max = 48.04 ± 0.11
χ²/ndf = 22.24
SNR = 1.43
r:1 c:2 m:0
Total events: 818811
Landau-Gauss fit, trigg.
L MPV = 38.89 ± 0.09
Max = 41.67 ± 0.10
χ²/ndf = 16.62
SNR = 1.56
r:1 c:3 m:0
Total events: 839534
Landau-Gauss fit, trigg.
L MPV = 41.69 ± 0.12
Max = 45.09 ± 0.13
χ²/ndf = 26.60
SNR = 1.40
r:0 c:0 m:0
Total events: 841334
Landau-Gauss fit, trigg.
L MPV = 41.17 ± 0.15
Max = 44.51 ± 0.16
χ²/ndf = 16.78
SNR = 1.66
r:0 c:1 m:0
Total events: 839995
Landau-Gauss fit, trigg.
L MPV = 43.28 ± 0.11
Max = 46.30 ± 0.12
χ²/ndf = 28.05
SNR = 1.58
r:0 c:2 m:0
Total events: 839658
Landau-Gauss fit, trigg.
L MPV = 42.23 ± 0.12
Max = 45.30 ± 0.13
χ²/ndf = 22.18
SNR = 1.67
r:0 c:3 m:0
Total events: 841272
Landau-Gauss fit, trigg.
L MPV = 41.98 ± 0.20
Max = 45.55 ± 0.22
χ²/ndf = 14.21
SNR = 1.39
counts
ADC (arb. units)

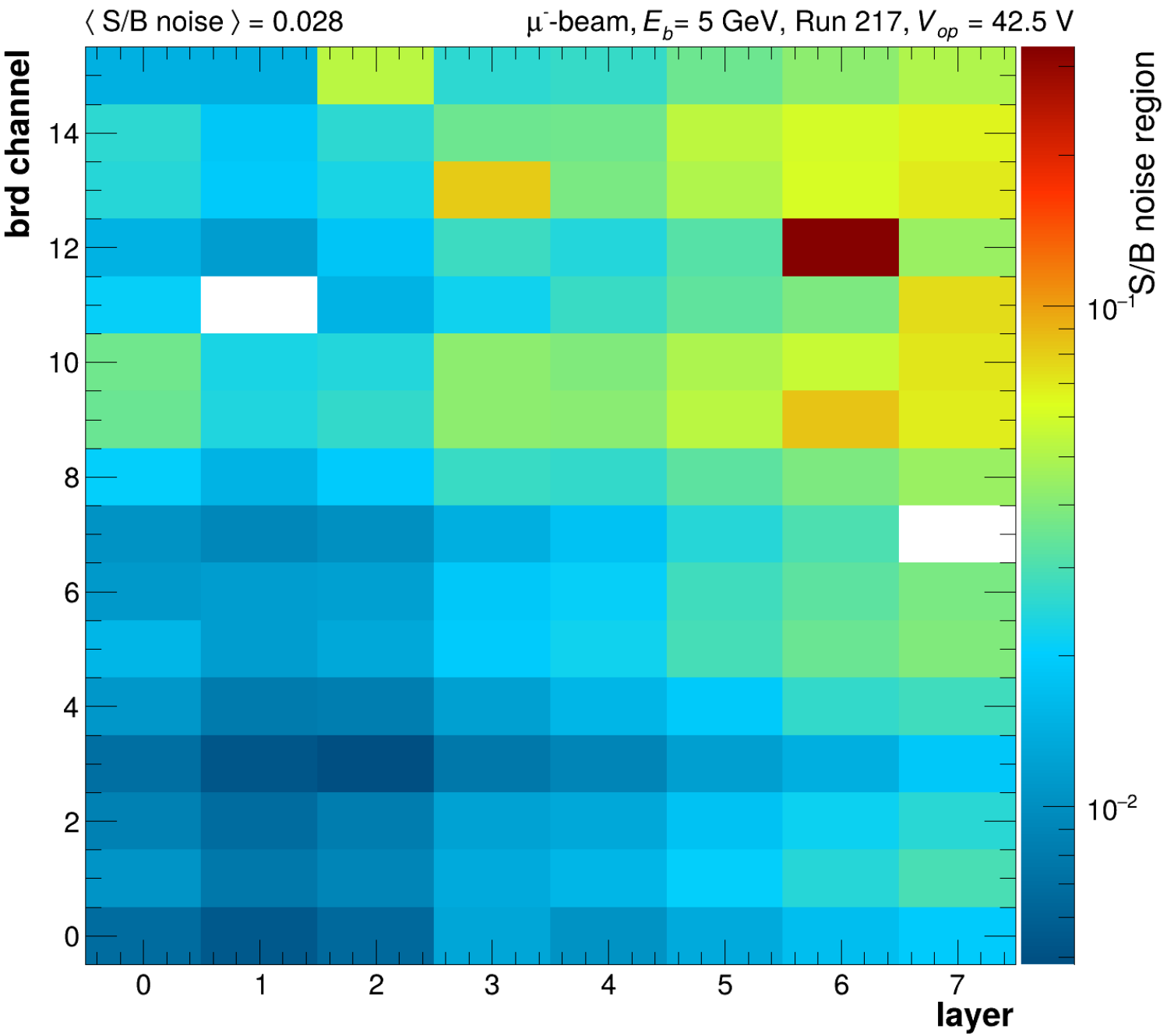

r:1 c:0 m:1
Total events: 136280
Landau-Gauss fit, trigg.
L MPV = 54.23 ± 0.21
Max = 58.05 ± 0.23
χ²/ndf = 36.96
SNR = 1.67
r:1 c:1 m:1
Total events: 592868
Landau-Gauss fit, trigg.
L MPV = 47.96 ± 0.13
Max = 51.62 ± 0.14
χ²/ndf = 24.89
SNR = 1.01
r:1 c:2 m:1
Total events: 585064
Landau-Gauss fit, trigg.
L MPV = 45.15 ± 0.12
Max = 48.49 ± 0.13
χ²/ndf = 35.72
SNR = 0.92
r:1 c:3 m:1 layer 6
Total events: 593388
Landau-Gauss fit, trigg.
L MPV = 45.91 ± 0.18
Max = 49.00 ± 0.19
χ²/ndf = 36.83
SNR = 1.18
μ⁻-beam, E_b = 5 GeV
Run 217, V_op = 42.5 V
r:0 c:0 m:1
Total events: 598018
Landau-Gauss fit, trigg.
L MPV = 43.18 ± 0.20
Max = 47.16 ± 0.22
χ²/ndf = 13.58
SNR = 1.14
r:0 c:1 m:1
Total events: 445848
Landau-Gauss fit, trigg.
L MPV = 54.14 ± 0.18
Max = 57.97 ± 0.19
χ²/ndf = 42.15
SNR = 1.41
r:0 c:2 m:1
Total events: 590196
Landau-Gauss fit, trigg.
L MPV = 47.15 ± 0.15
Max = 50.70 ± 0.16
χ²/ndf = 22.45
SNR = 1.19
r:0 c:3 m:1
Total events: 593453
Landau-Gauss fit, trigg.
L MPV = 46.57 ± 0.22
Max = 50.30 ± 0.24
χ²/ndf = 14.06
SNR = 0.88
r:1 c:0 m:0
Total events: 639187
Landau-Gauss fit, trigg.
L MPV = 46.68 ± 0.15
Max = 49.95 ± 0.16
χ²/ndf = 28.66
SNR = 7.64
r:1 c:1 m:0
Total events: 840981
Landau-Gauss fit, trigg.
L MPV = 48.23 ± 0.15
Max = 52.00 ± 0.16
χ²/ndf = 33.45
SNR = 1.07
r:1 c:2 m:0
Total events: 839988
Landau-Gauss fit, trigg.
L MPV = 48.10 ± 0.13
Max = 51.61 ± 0.14
χ²/ndf = 34.93
SNR = 1.68
r:1 c:3 m:0
Total events: 841265
Landau-Gauss fit, trigg.
L MPV = 38.54 ± 0.27
Max = 38.27 ± 0.27
χ²/ndf = 39.94
SNR = 0.97
r:0 c:0 m:0
Total events: 841656
Landau-Gauss fit, trigg.
L MPV = 42.74 ± 0.23
Max = 46.37 ± 0.24
χ²/ndf = 10.87
SNR = 1.17
r:0 c:1 m:0
Total events: 775468
Landau-Gauss fit, trigg.
L MPV = 44.77 ± 0.14
Max = 48.04 ± 0.15
χ²/ndf = 28.25
SNR = 2.16
r:0 c:2 m:0
Total events: 840578
Landau-Gauss fit, trigg.
L MPV = 45.63 ± 0.15
Max = 48.92 ± 0.16
χ²/ndf = 14.20
SNR = 1.27
r:0 c:3 m:0
Total events: 841394
Landau-Gauss fit, trigg.
L MPV = 44.91 ± 0.28
Max = 44.72 ± 0.28
χ²/ndf = 21.97
SNR = 1.10
counts
ADC (arb. units)

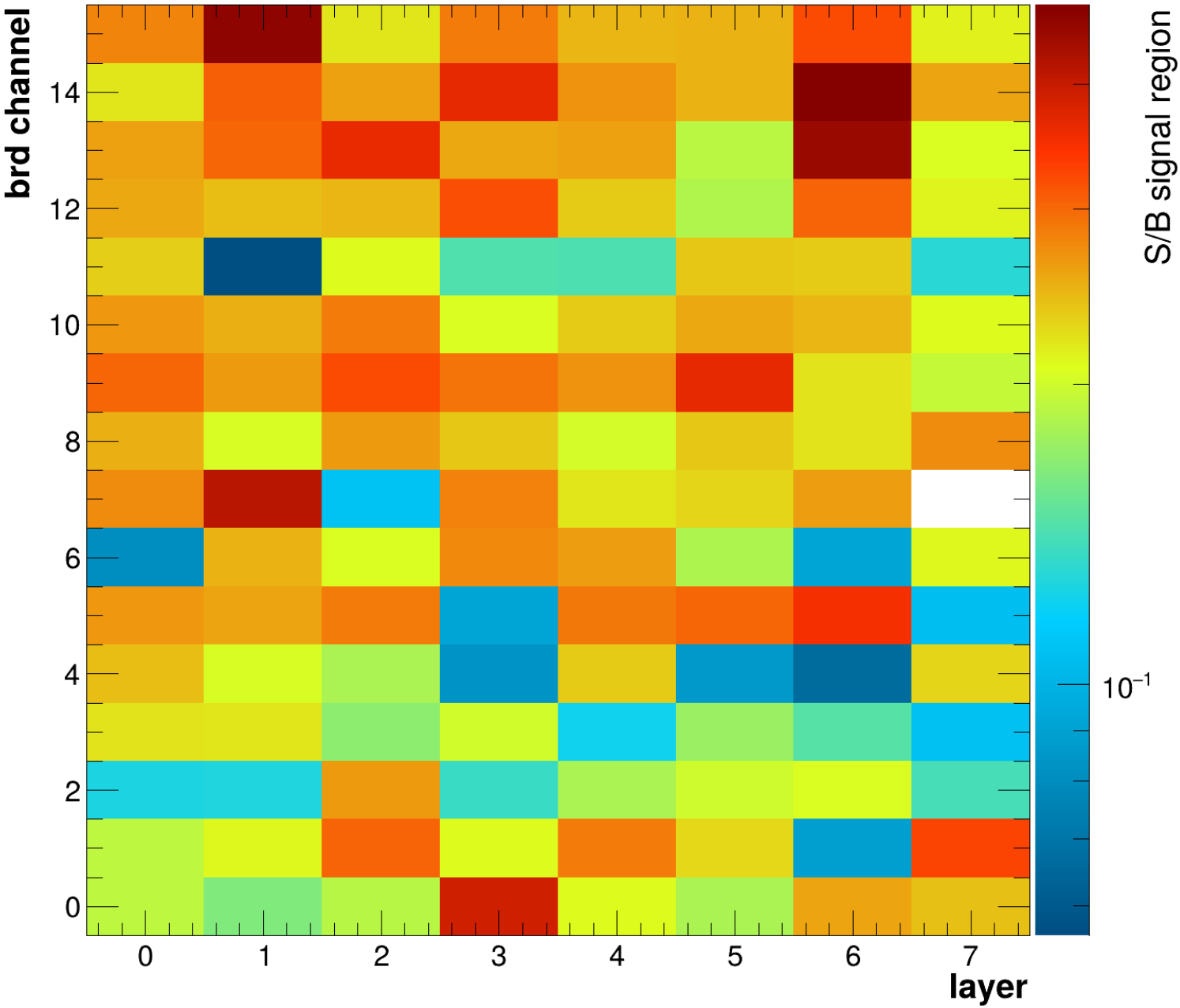

r:1 c:0 m:1
Total events: 598336
Landau-Gauss fit, trigg.
L MPV = 44.66 ± 0.28
Max = 49.00 ± 0.30
χ²/ndf = 14.95
SNR = 0.97
r:1 c:1 m:1
Total events: 589880
Landau-Gauss fit, trigg.
L MPV = 51.48 ± 0.16
Max = 55.17 ± 0.17
χ²/ndf = 26.75
SNR = 1.04
r:1 c:2 m:1
Total events: 585334
Landau-Gauss fit, trigg.
L MPV = 49.04 ± 0.18
Max = 52.60 ± 0.19
χ²/ndf = 37.98
SNR = 0.86
r:1 c:3 m:1 layer 7
Total events: 580293
Landau-Gauss fit, trigg.
L MPV = 50.47 ± 0.20
Max = 54.27 ± 0.21
χ²/ndf = 21.04
SNR = 0.93
μ-beam, E_b = 5 GeV
Run 217, V_{op} = 42.5 V
r:0 c:0 m:1
Total events: 594563
Landau-Gauss fit, trigg.
L MPV = 35.88 ± 0.07
Max = 36.85 ± 0.07
χ²/ndf = 37.43
SNR = 0.81
r:0 c:1 m:1
Total events: 589508
Landau-Gauss fit, trigg.
L MPV = 53.72 ± 0.19
Max = 57.56 ± 0.20
χ²/ndf = 25.94
SNR = 0.96
r:0 c:2 m:1
Total events: 532844
Landau-Gauss fit, trigg.
L MPV = 53.24 ± 0.20
Max = 57.17 ± 0.21
χ²/ndf = 30.31
SNR = 1.00
r:0 c:3 m:1
Total events: 375088
Landau-Gauss fit, trigg.
L MPV = 61.60 ± 0.05
Max = 60.49 ± 0.05
χ²/ndf = 21.26
SNR = 1.11
r:1 c:0 m:0
Total events: 841705
Landau-Gauss fit, trigg.
L MPV = 44.05 ± 0.24
Max = 48.28 ± 0.26
χ²/ndf = 21.35
SNR = 0.83
r:1 c:1 m:0
Total events: 840495
Landau-Gauss fit, trigg.
L MPV = 51.78 ± 0.18
Max = 55.72 ± 0.19
χ²/ndf = 23.83
SNR = 1.40
r:1 c:2 m:0
Total events: 840356
Landau-Gauss fit, trigg.
L MPV = 53.12 ± 0.20
Max = 56.93 ± 0.21
χ²/ndf = 28.66
SNR = 0.85
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 841667
Landau-Gauss fit, trigg.
L MPV = 47.11 ± 0.24
Max = 50.77 ± 0.26
χ²/ndf = 11.32
SNR = 0.91
r:0 c:1 m:0
Total events: 840576
Landau-Gauss fit, trigg.
L MPV = 50.52 ± 0.20
Max = 54.21 ± 0.22
χ²/ndf = 15.27
SNR = 0.81
r:0 c:2 m:0
Total events: 819388
Landau-Gauss fit, trigg.
L MPV = 48.83 ± 0.26
Max = 52.41 ± 0.28
χ²/ndf = 32.96
SNR = 1.09
r:0 c:3 m:0
Total events: 742161
Landau-Gauss fit, trigg.
L MPV = 49.44 ± 0.28
Max = 53.21 ± 0.31
χ²/ndf = 12.45
SNR = 0.83
counts
ADC (arb. units)

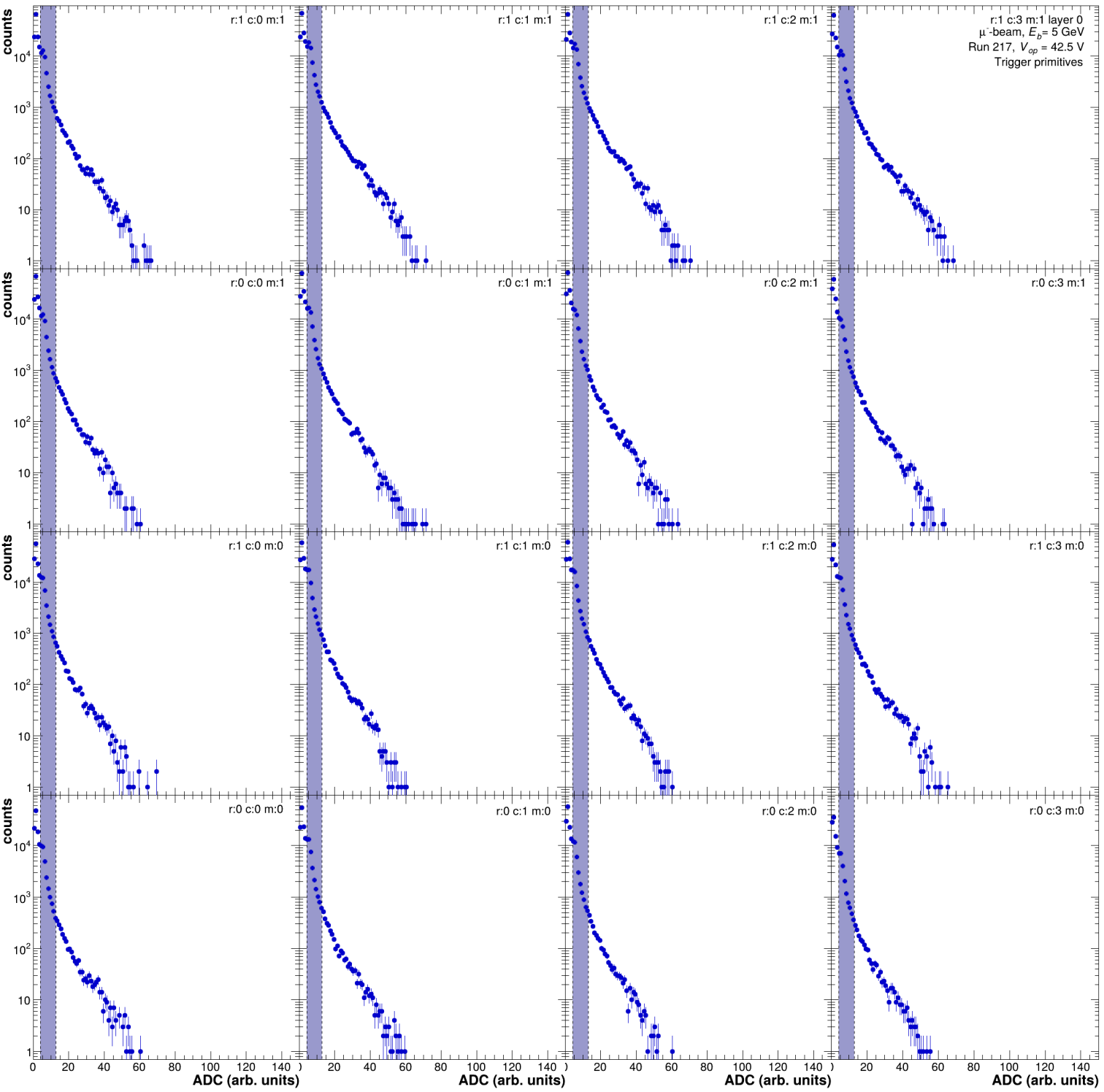

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
×10³
Num triggers
500
450
400
350
300
250
200
Y (row)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
X (col)
−0.5
0
0.5
1
1.5
2
2.5
3
3.5

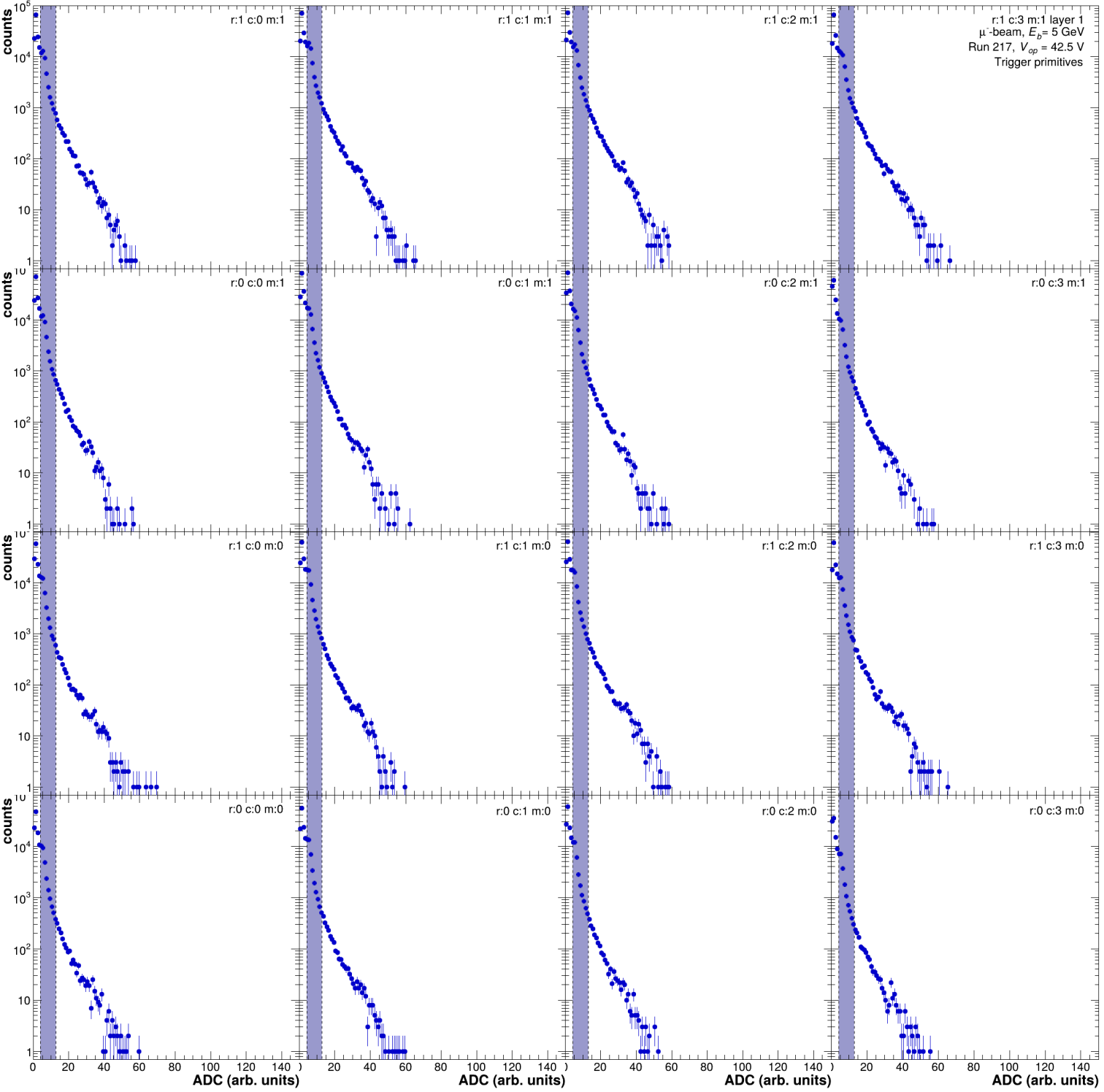

evt. collected = 842041
μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

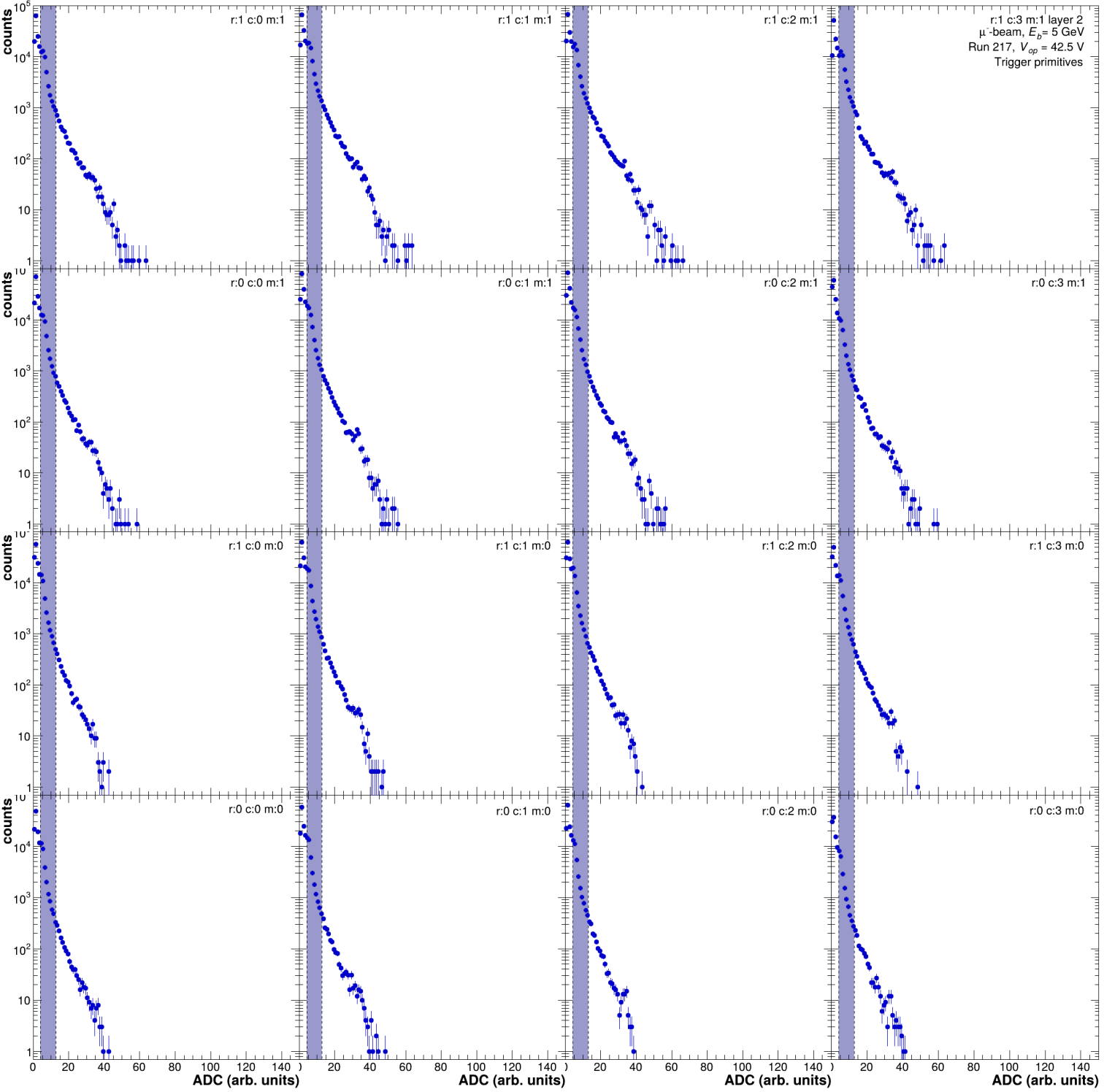

μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
layers
0
1
2
3
4
5
6
7
SNR
2
4
6
8
10
12
14
16
18

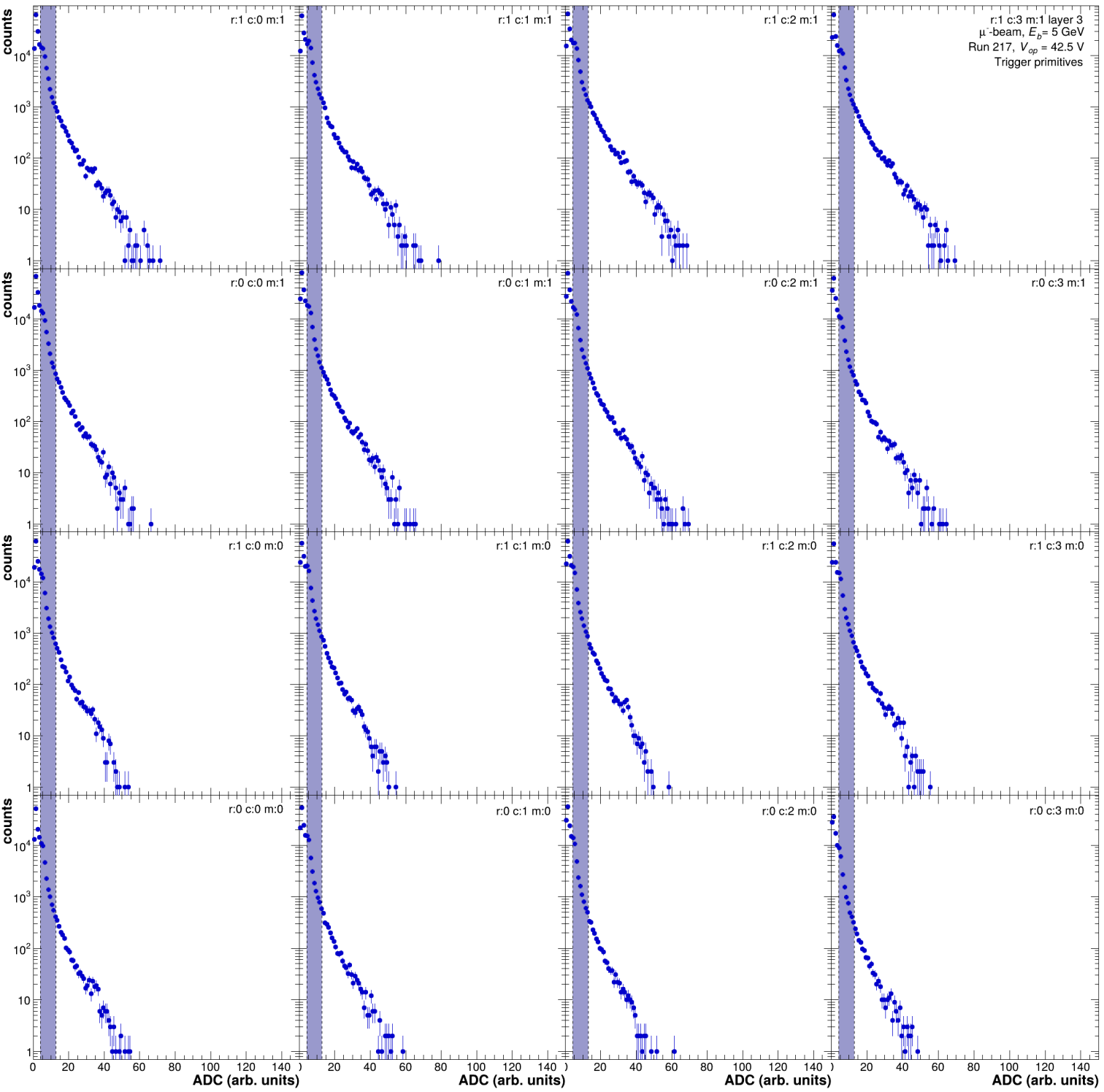

⟨ S/B noise ⟩ = 0.028
μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

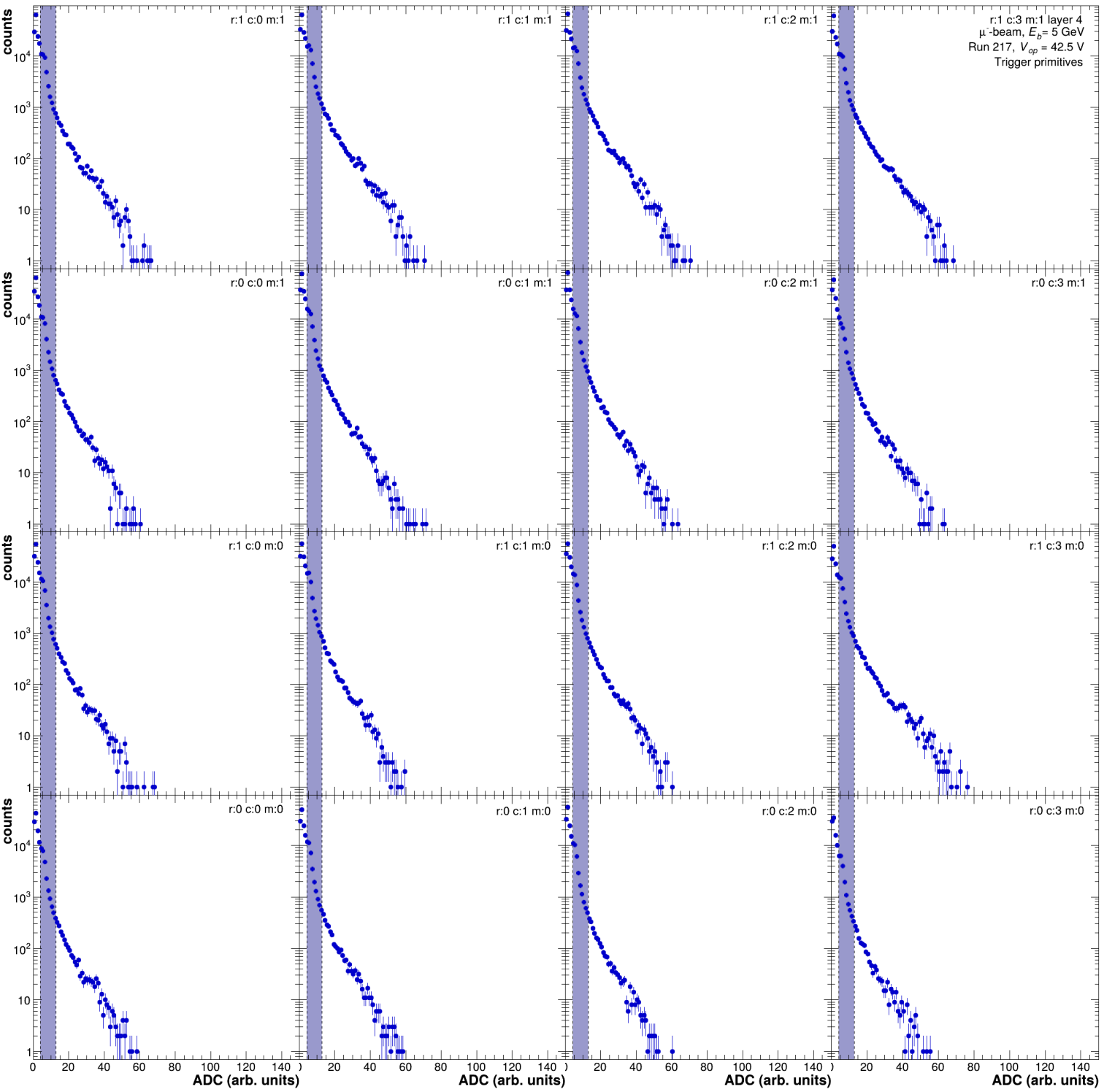

⟨ S/B signal ⟩ = 0.231
μ⁻-beam, E_b= 5 GeV, Run 217, V_{op} = 42.5 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

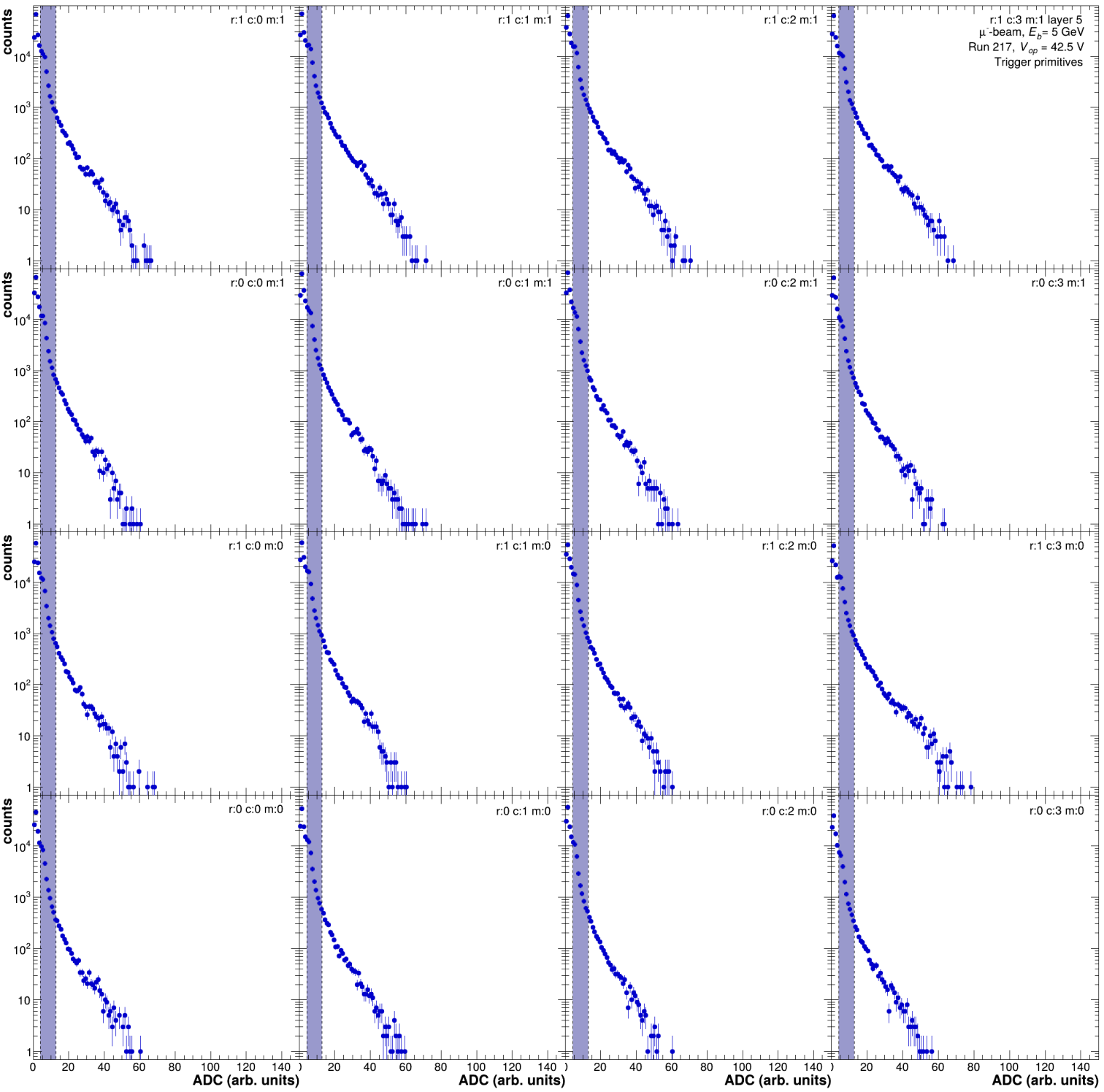

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

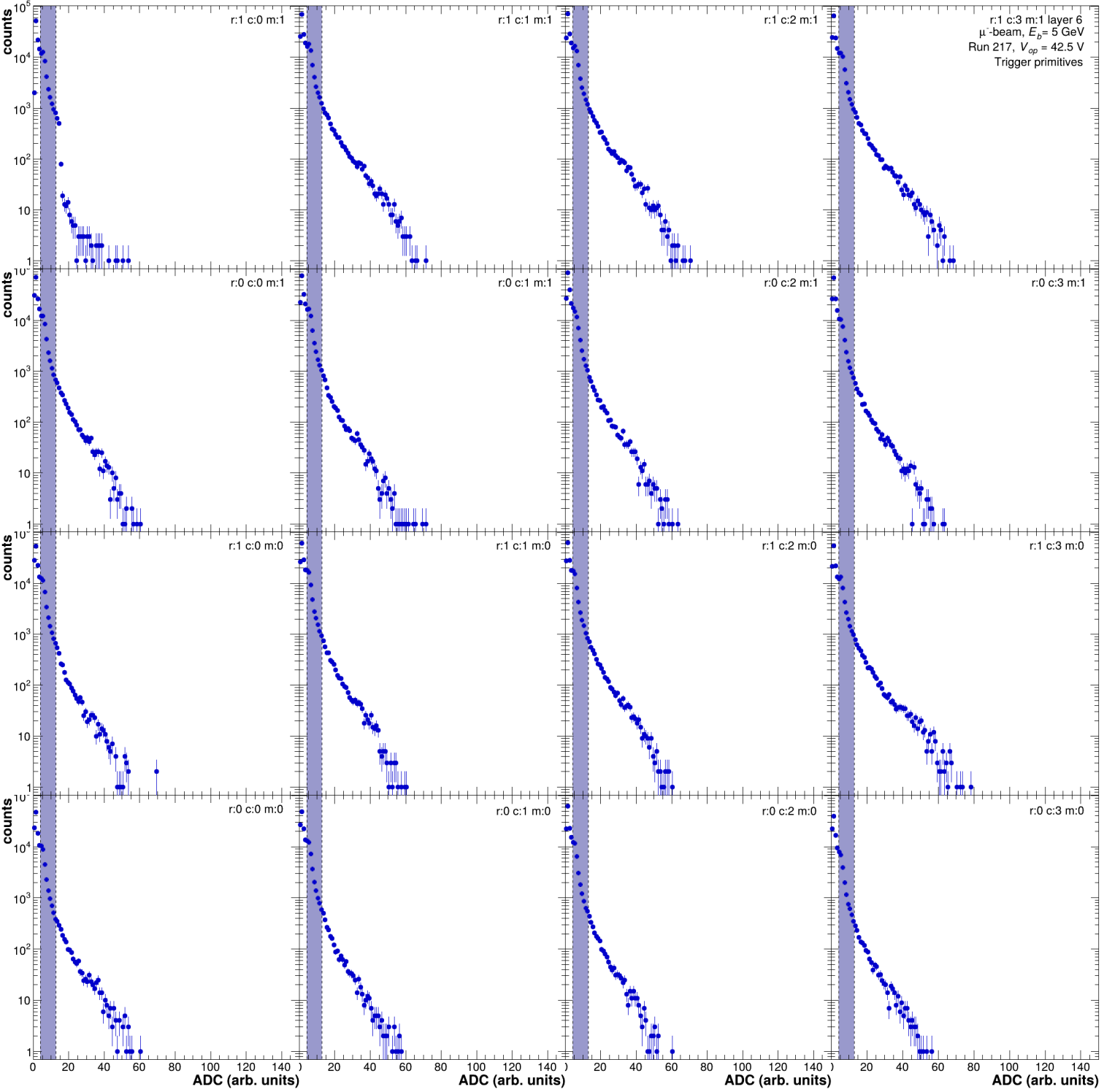

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

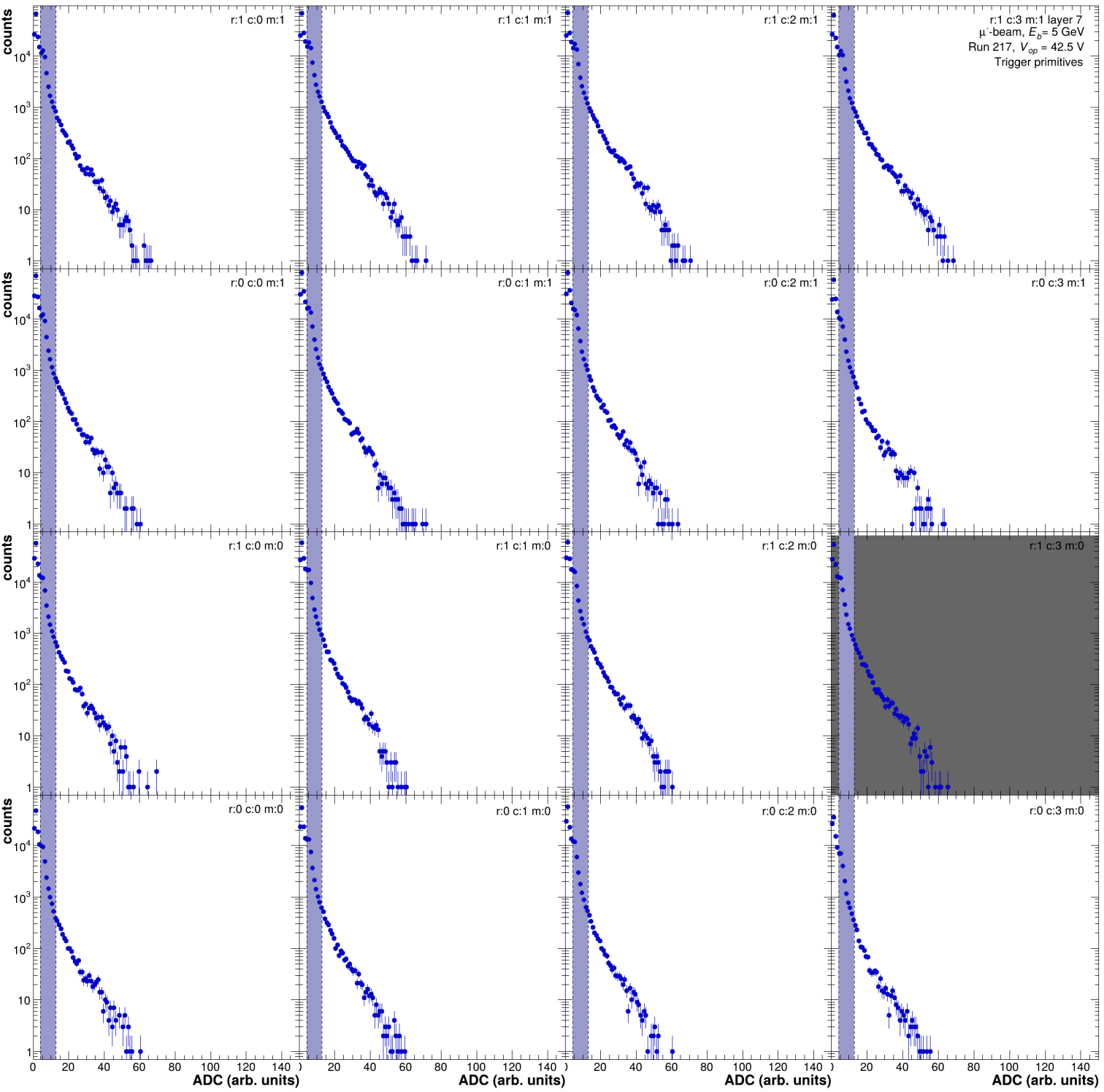

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 217, V_{op} = 42.5 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)